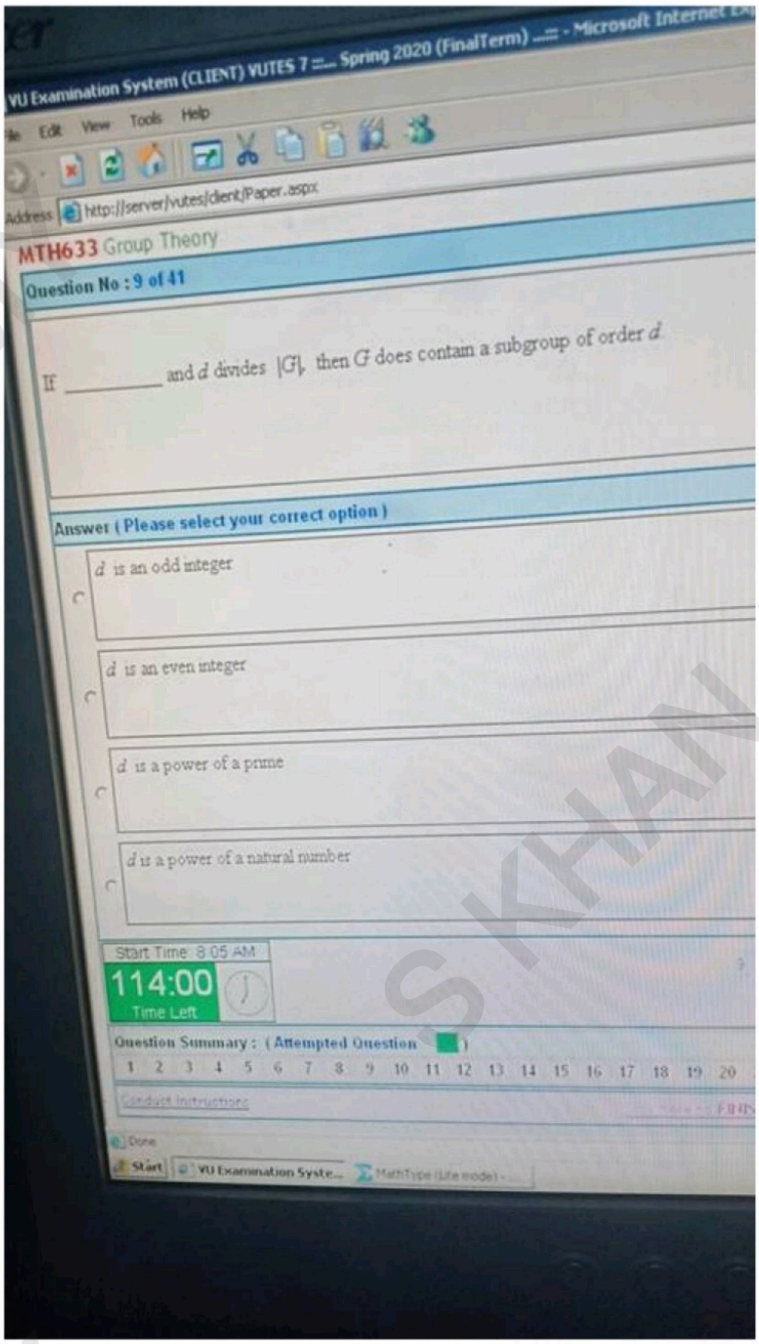

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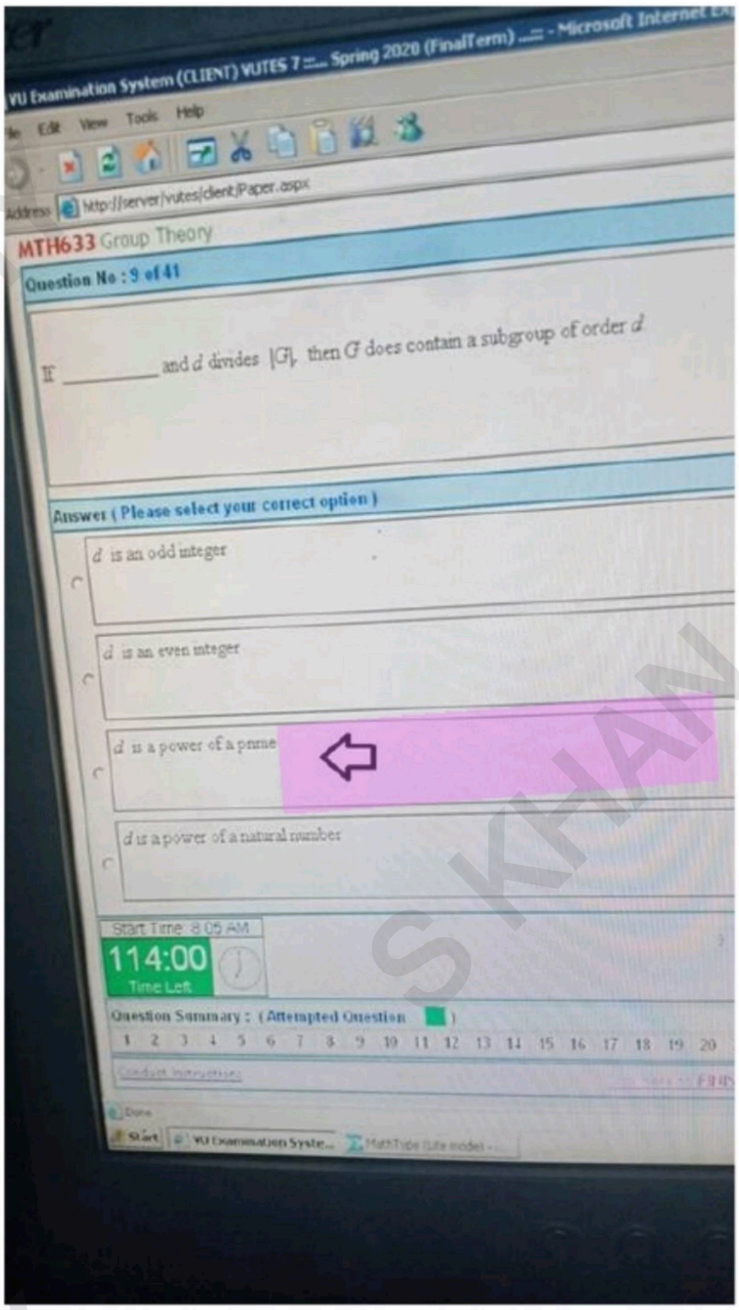

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Sylow Theorems

The Sylow theorems give a weak converse. Namely, they show that if d is a power of a prime and d divides |G|, then G does contain a subgroup of order d.

Note that 6 is not a power of a prime. The Sylow theorems also give some information concerning the number of such subgroups and their relationship to each other.

We will see that these theorems are very useful in studying finite nonabelian groups.

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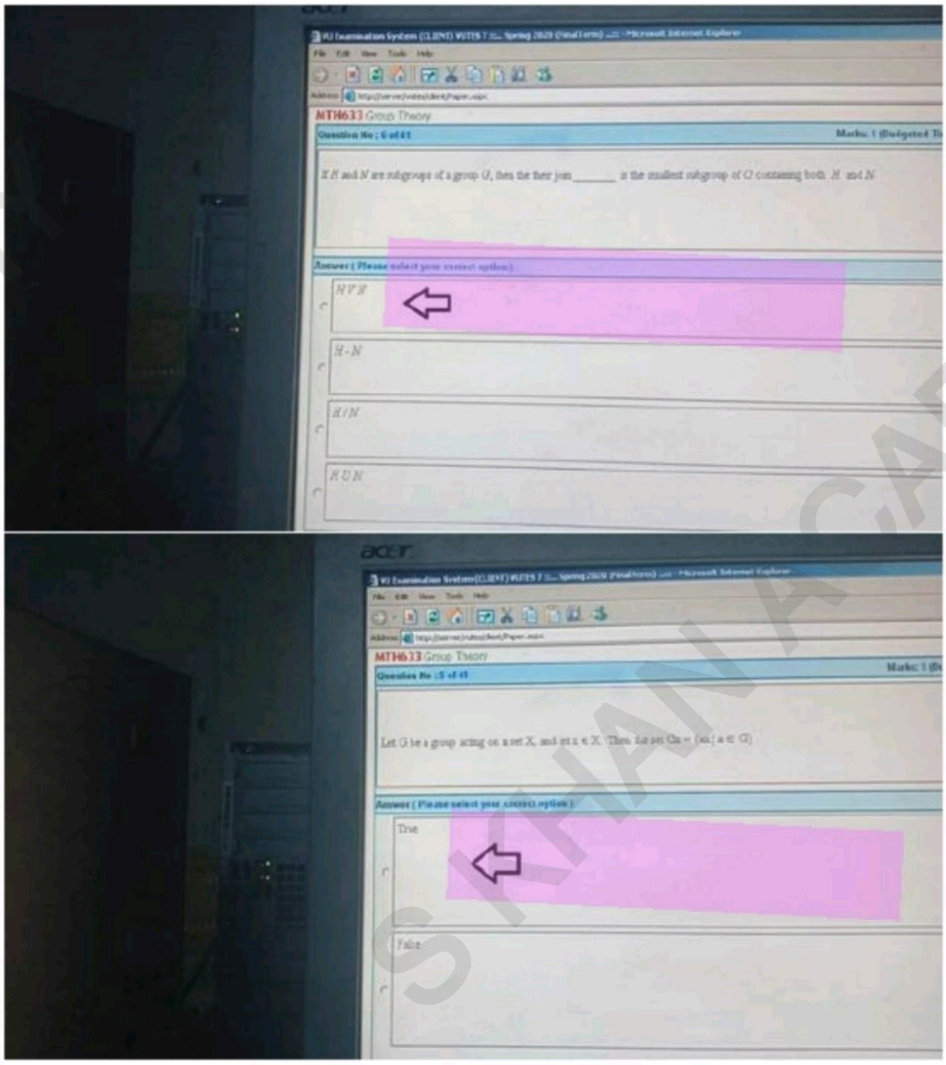

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rphism Theorems

If H and N are subgroups of a group G, then we let

$HN=\{hn \mid h \in H, n \in N\}$.

We define the **join** $H \vee N$ of H and N as the intersection of all subgroups of G that contain HN; thus $H \vee N$ is the smallest subgroup of G containing HN.

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morphism Theorems

Of course H V N is also the smallest subgroup of G containing both H and N, since any such subgroup must contain HN. In general, HN need not be a subgroup of G.

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Orbits

Definition

Let G be a group acting on a set X, and let $x \in X$. Then the set

$Gx = \{ax \mid a \in G\}$

is called the orbit of x in G.

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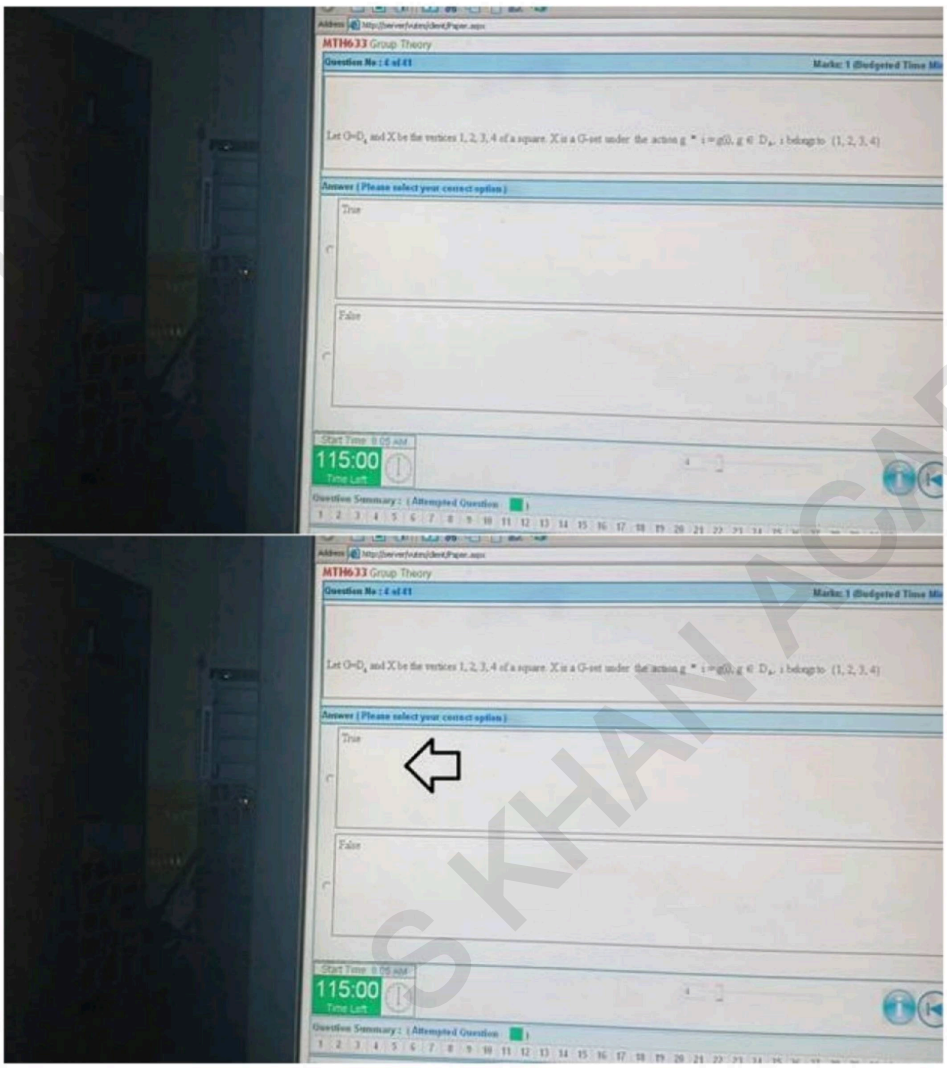

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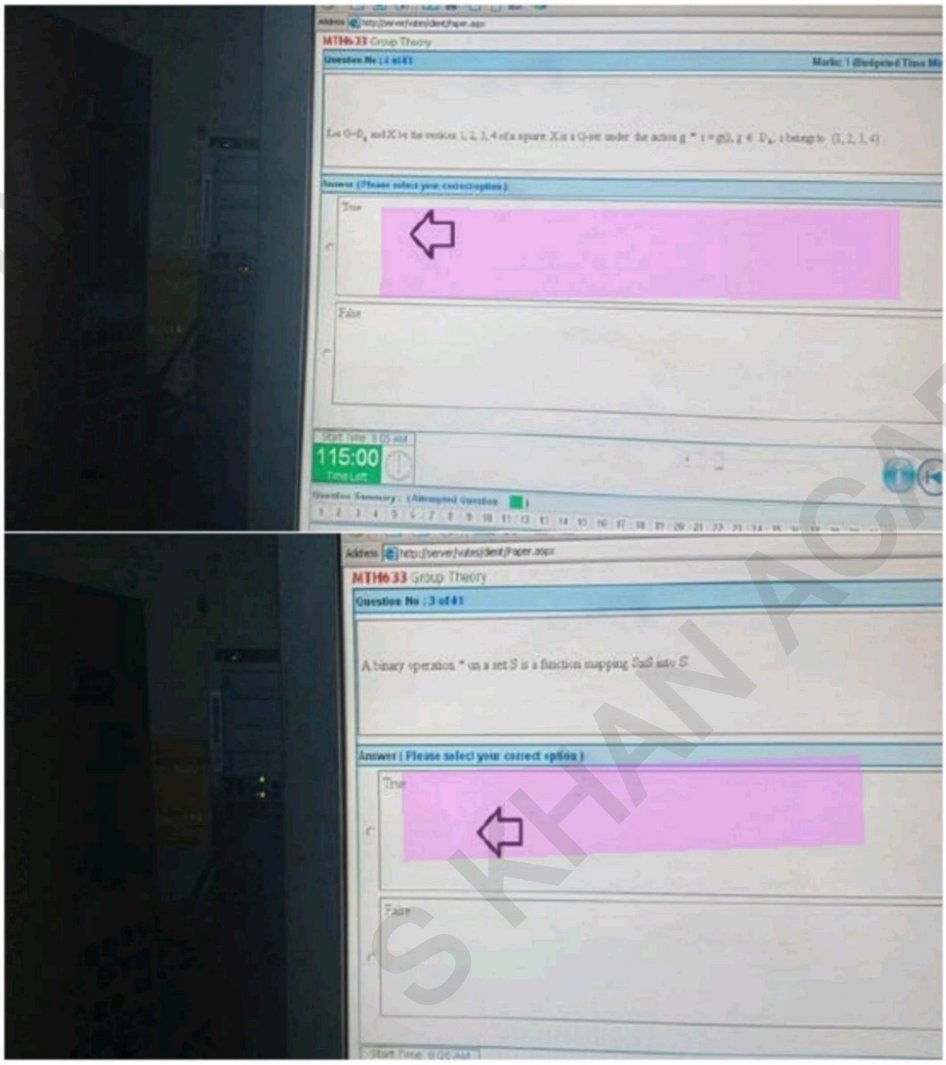

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p Action on a Set

Example

Let $G=D_4$ and X be the vertices 1, 2, 3, 4 of a square. X is a G-set under the action

$g * i = g(i)$, $g \in D_4$,

$i \in \{1, 2, 3, 4\}$.

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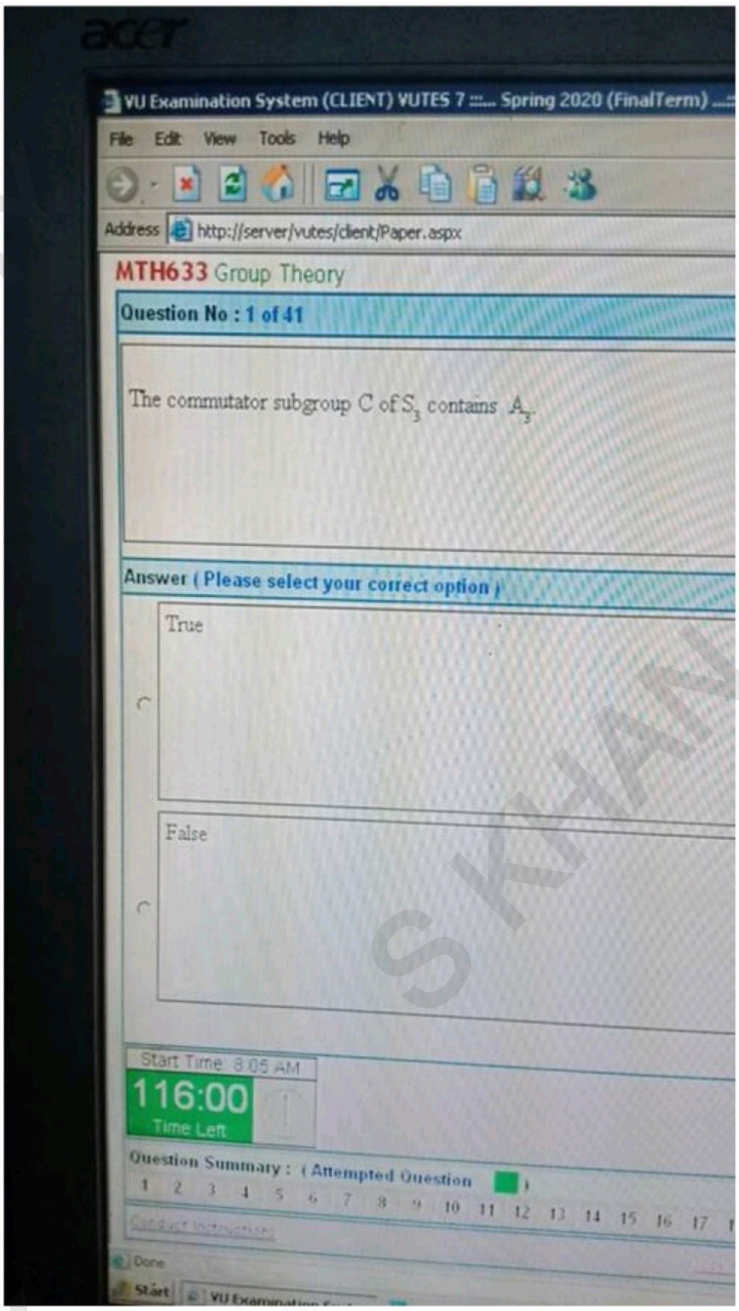

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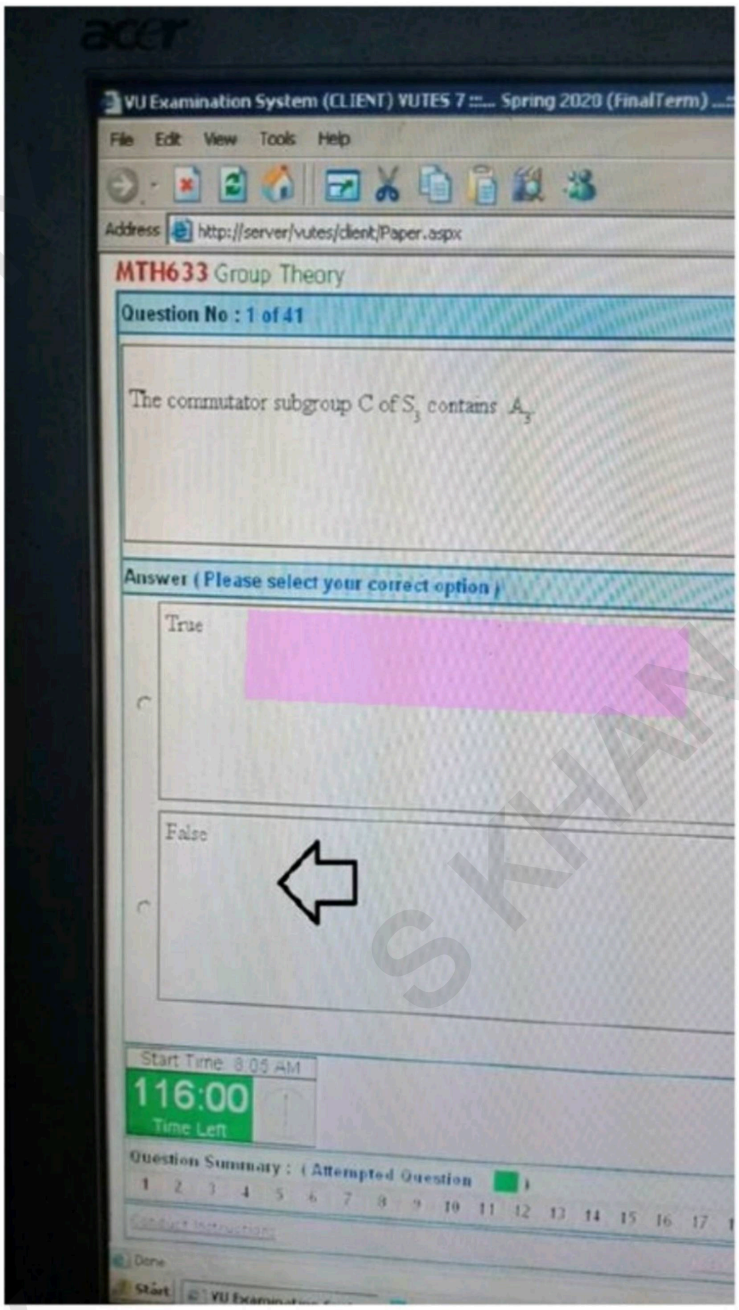

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mutator Subgroup

Thus the commutator subgroup C of S_3 contains A_3. Since A_3 is a normal subgroup of S_3 and

S_3/A_3 is abelian, above theorem shows that $C=A_3$.

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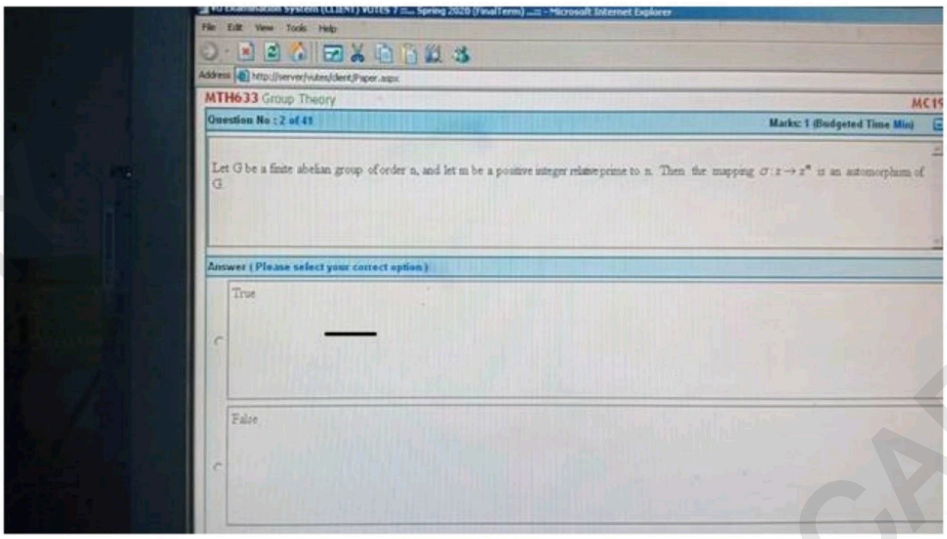

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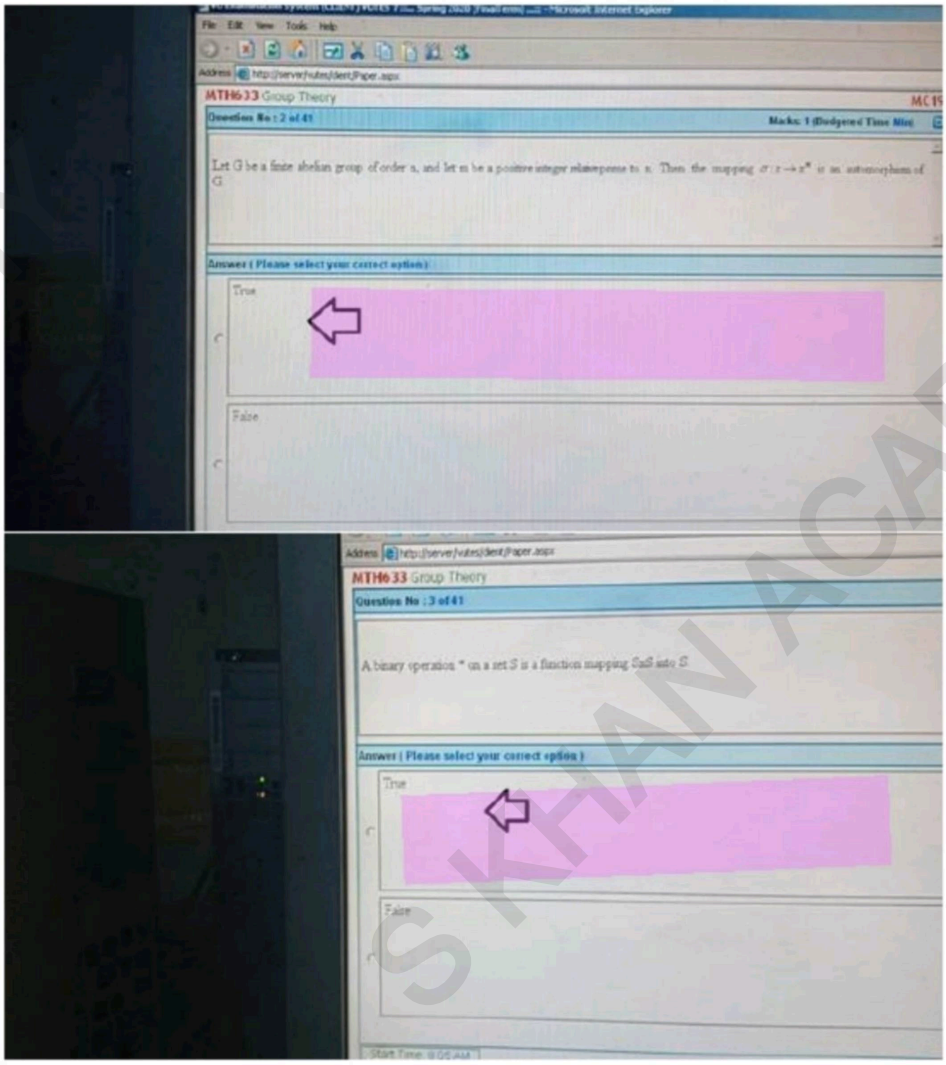

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Example

Let G be a finite abelian group of order n, and let m be a positive integer relative prime to n. Then the mapping $\sigma: x \rightarrow x^m$ is an automorphism of G.

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Definition

A binary operation $*$ on a set S is a function mapping S x S into S.

For each $(a, b) \in S \times S$, we will denote the element $*((a, b))$ of S by $a*b$.

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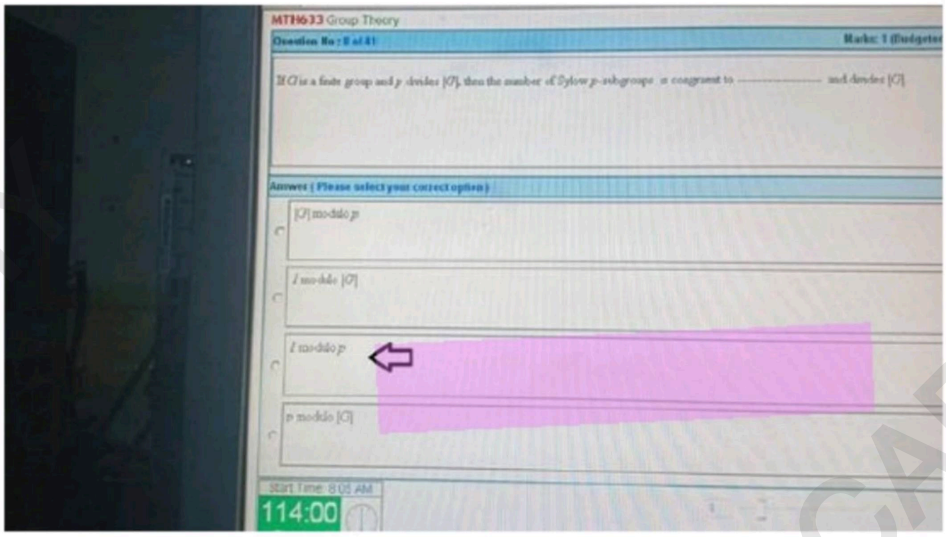

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Theorem

If G is a finite group and p divides |G|, then the number of Sylow p-subgroups is congruent to 1 modulo p and divides |G|.

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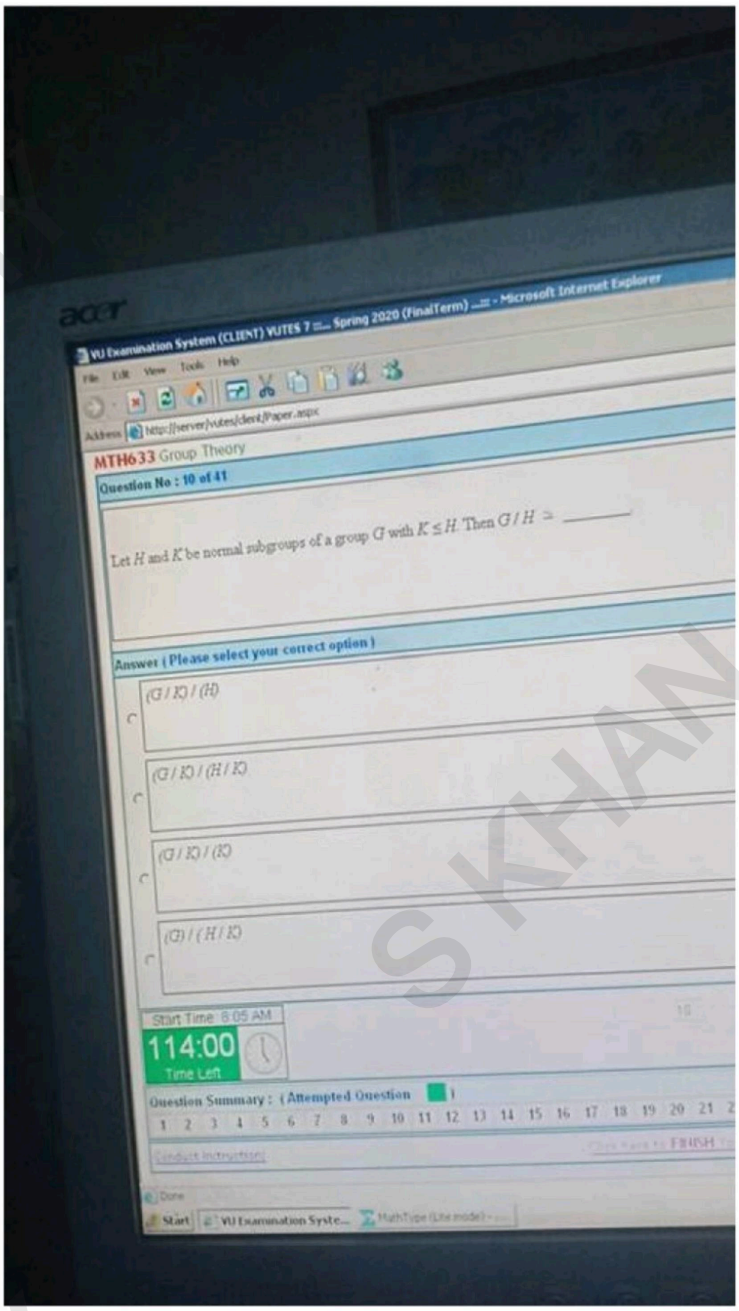

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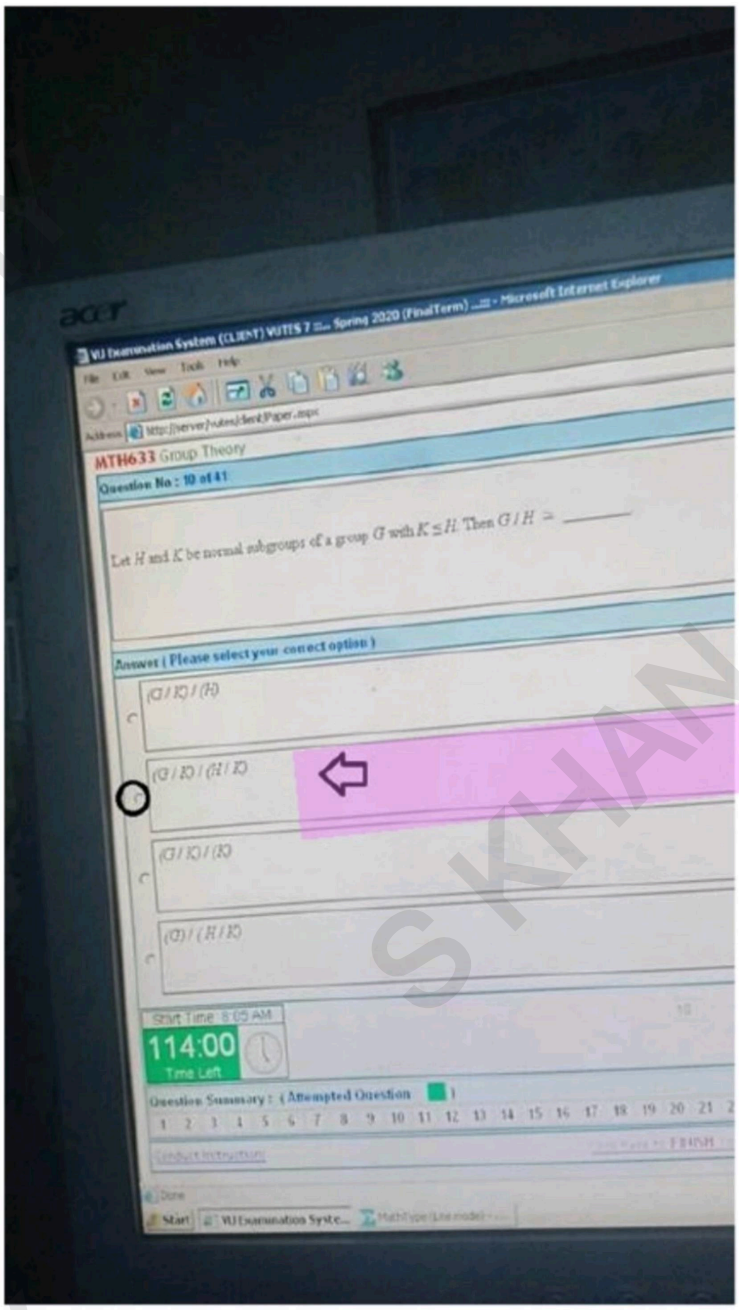

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morphism Theorem

Theorem

Let H and K be normal subgroups of a group G with $K \leq H$.

Then $G/H \simeq (G/K)/(H/K)$.

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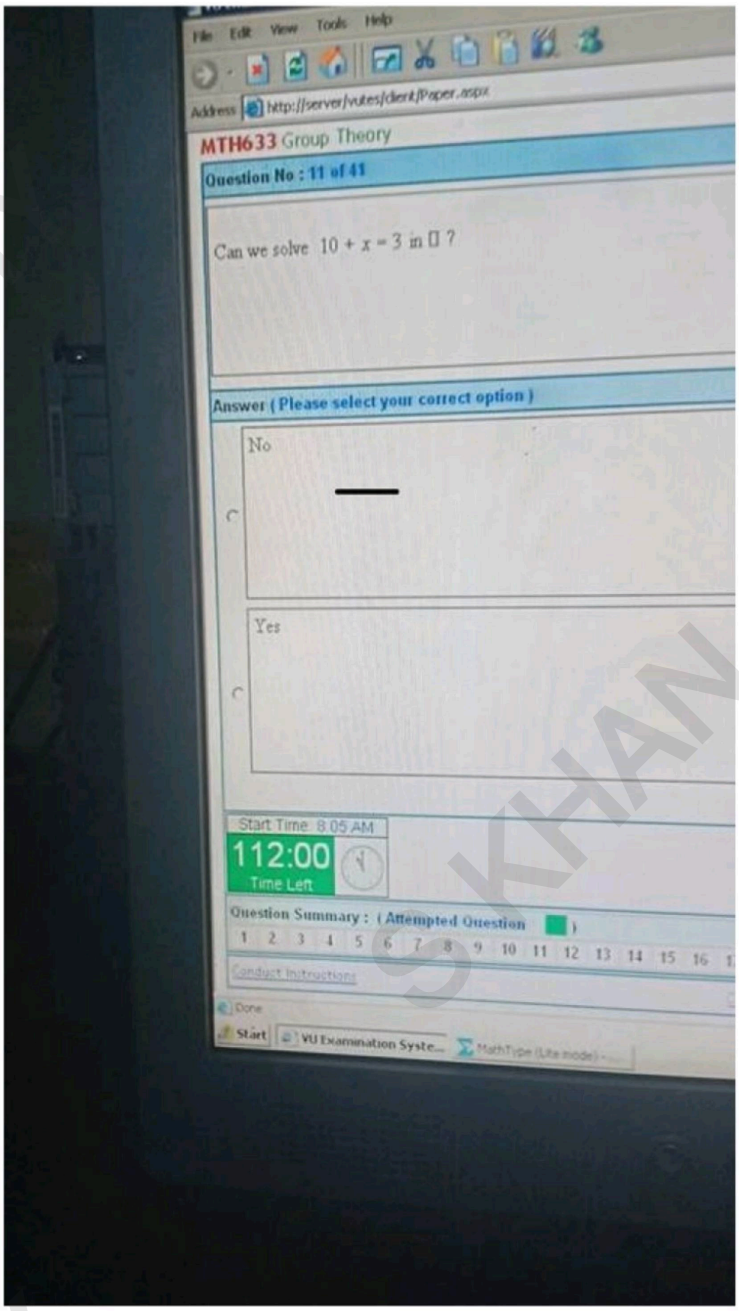

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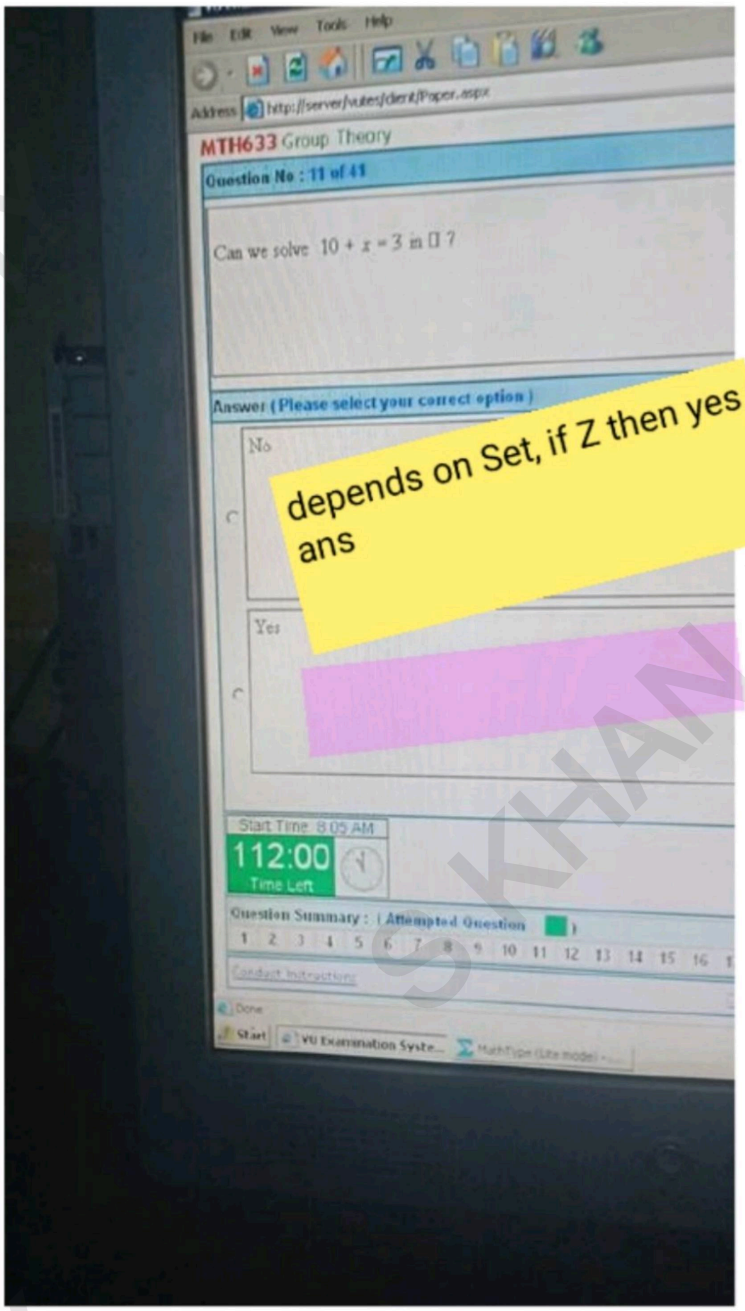

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Isomorphic Binary Structures

Example

- We prove that the binary structures $\langle \mathbb{Q}, + \rangle$ and $\langle \mathbb{Z}, + \rangle$ under the usual addition are not isomorphic.
- Both $\mathbb{Q}$ and $\mathbb{Z}$ have cardinality $\aleph_0$, so there are lots of one-to-one functions mapping $\mathbb{Q}$ onto $\mathbb{Z}$.
- The equation $x + x = c$ has a solution x for all $c \in \mathbb{Q}$ but this is not the case in $\mathbb{Z}$.

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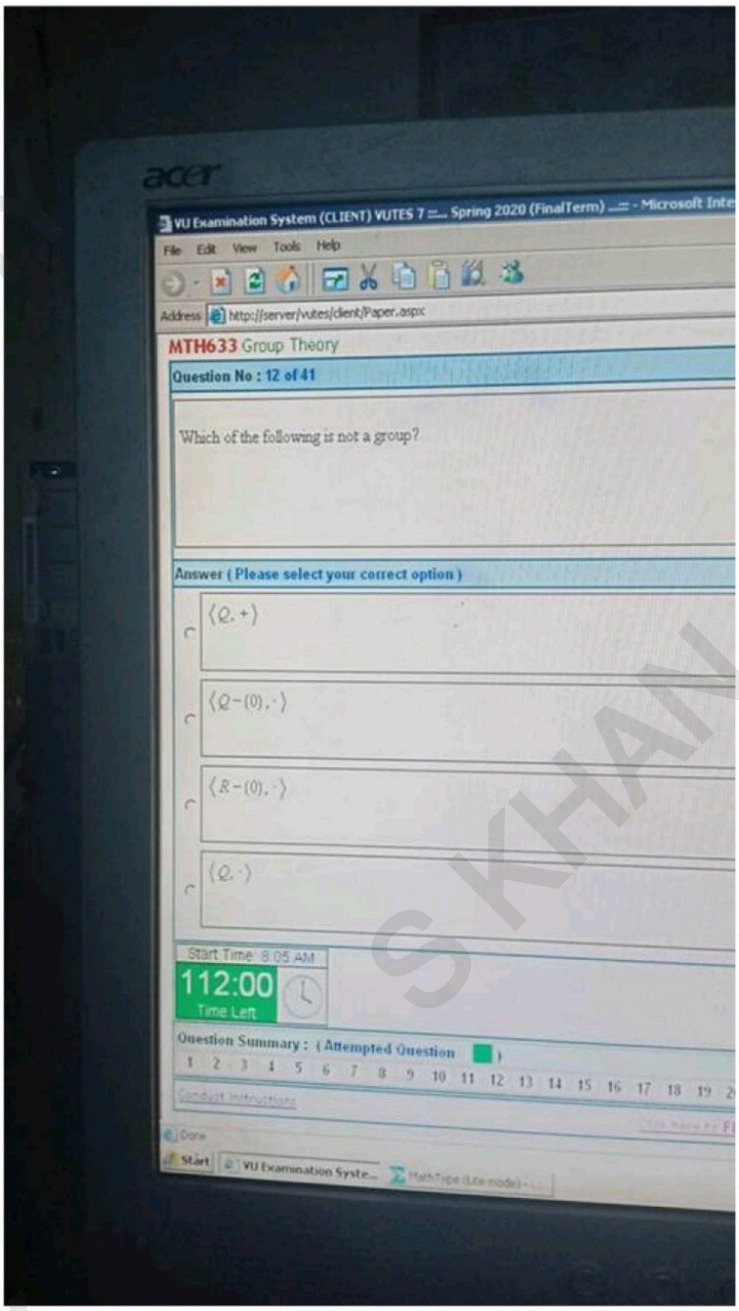

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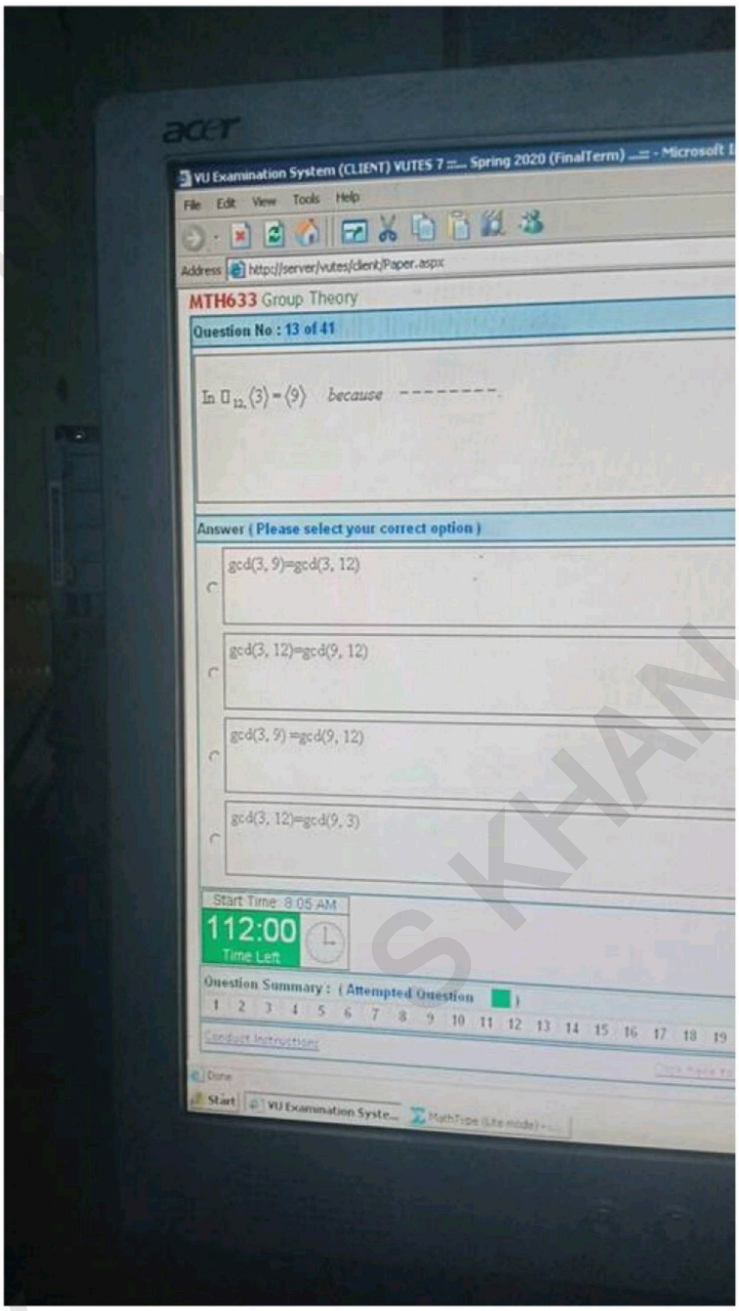

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Subgroups of Finite Cyclic Group

Example

using additive notation, consider in $\mathbb{Z}_{12}$, with the generator a=1.

- 3 = 3·1, gcd(3, 12)=3, so $\langle 3 \rangle$ has 12/3=4 elements. $\langle 3 \rangle$={0, 3, 6, 9}
- Furthermore, $\langle 3 \rangle = \langle 9 \rangle$ since gcd(3, 12)=gcd(9, 12).

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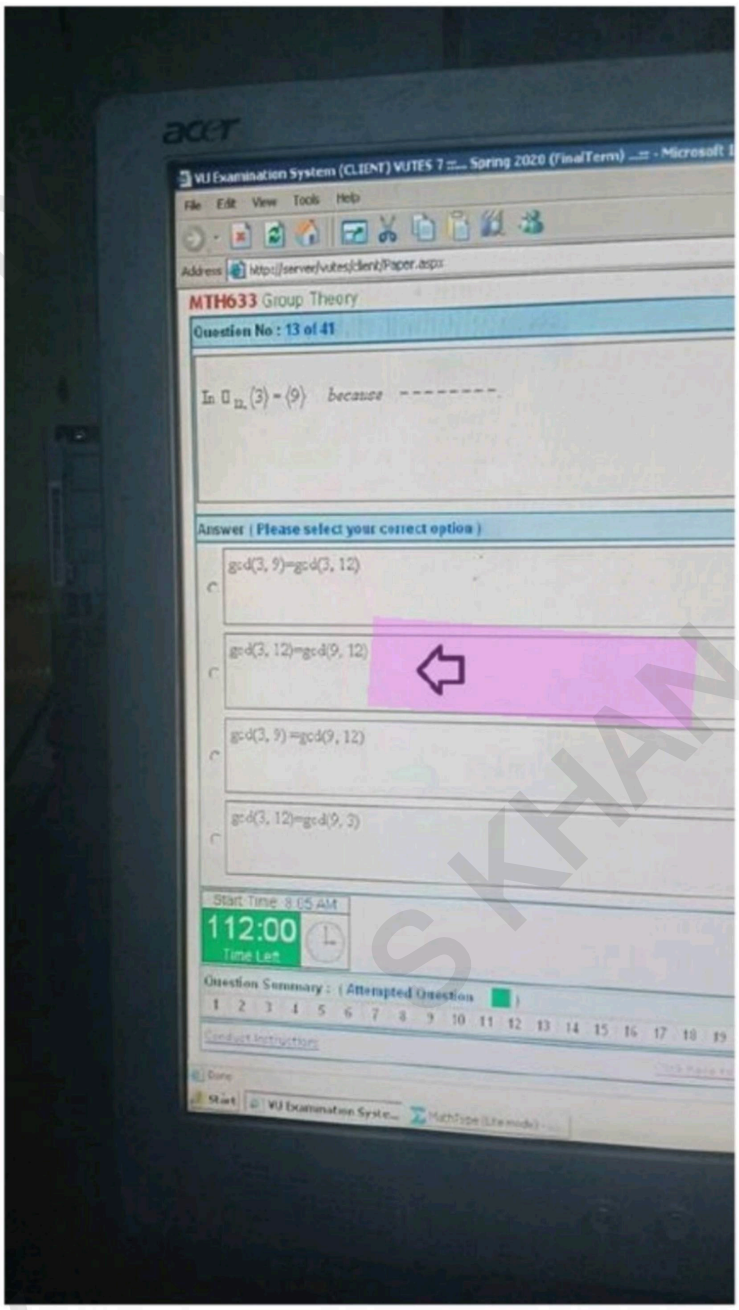

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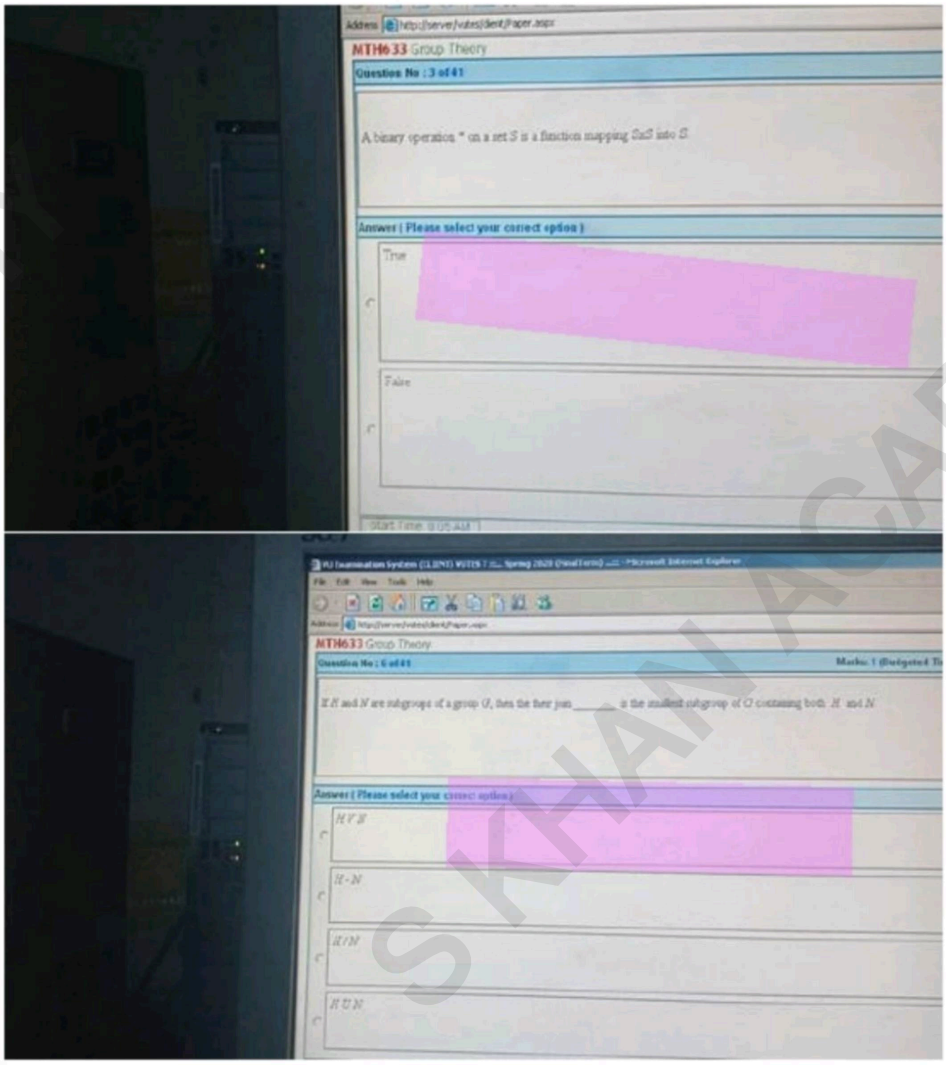

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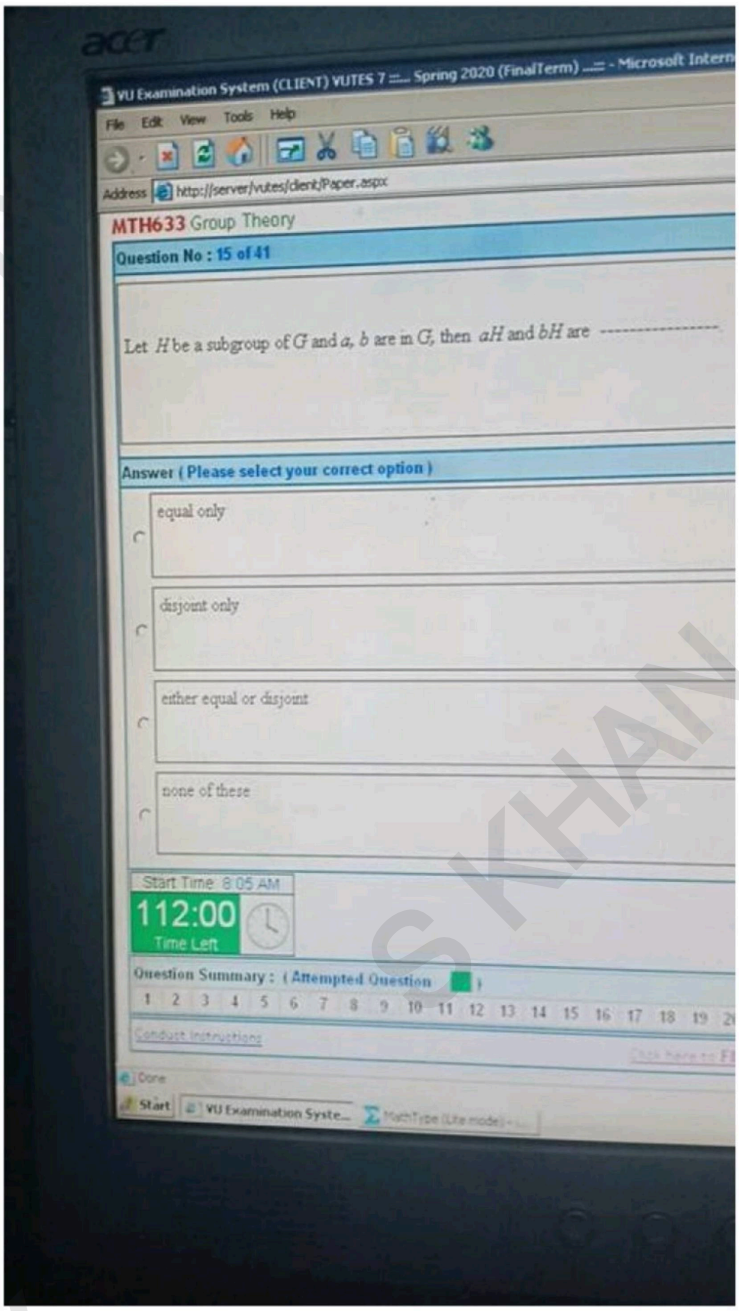

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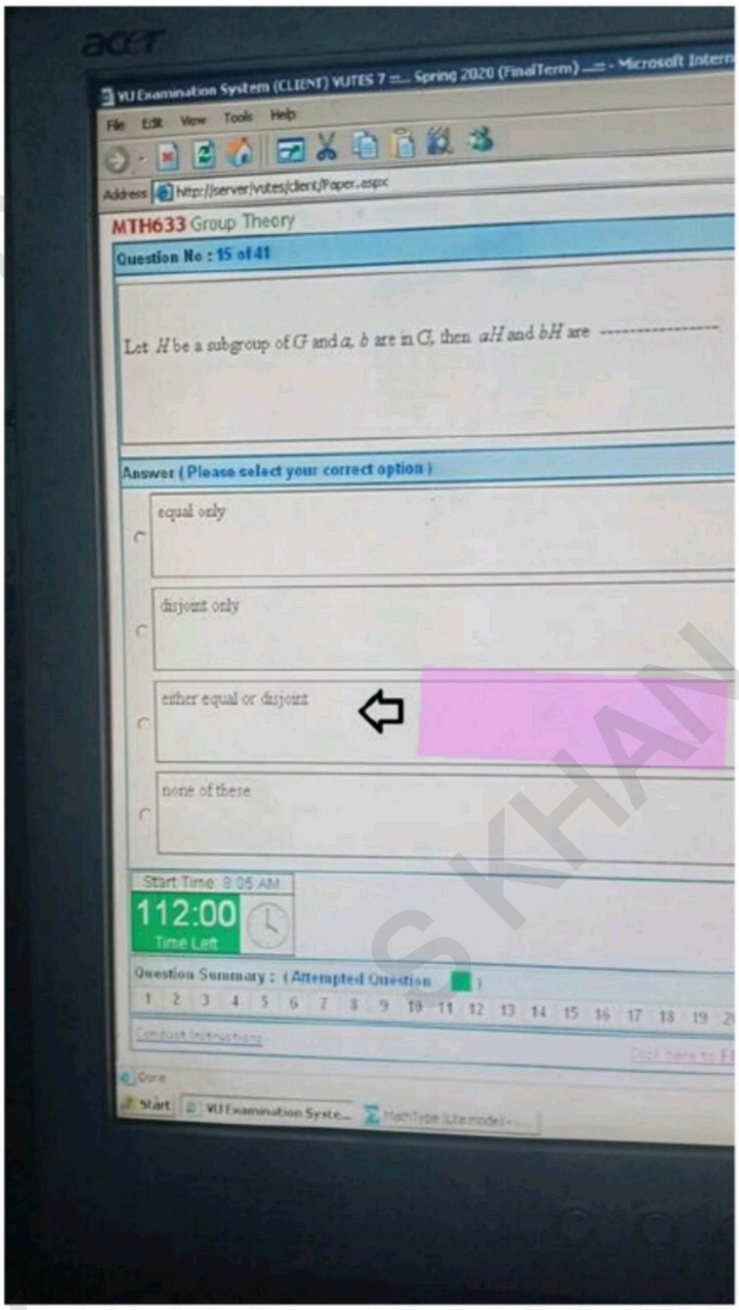

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Properties of Cosets

Proposition

Let H be a subgroup of G, and a,b in G.

3. $aH = bH$ iff a belongs to bH
4. aH and bH are either equal or disjoint
5. $aH = bH$ iff $a^{-1}b$ belongs to H

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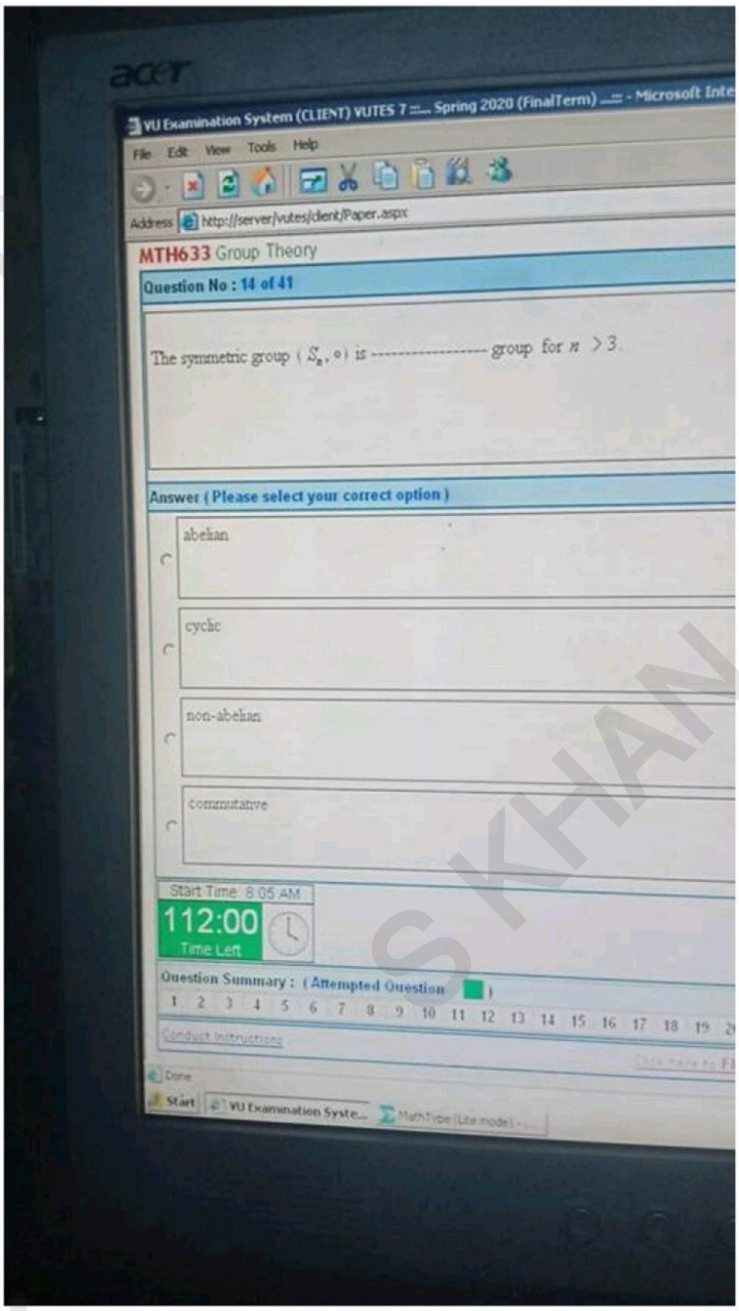

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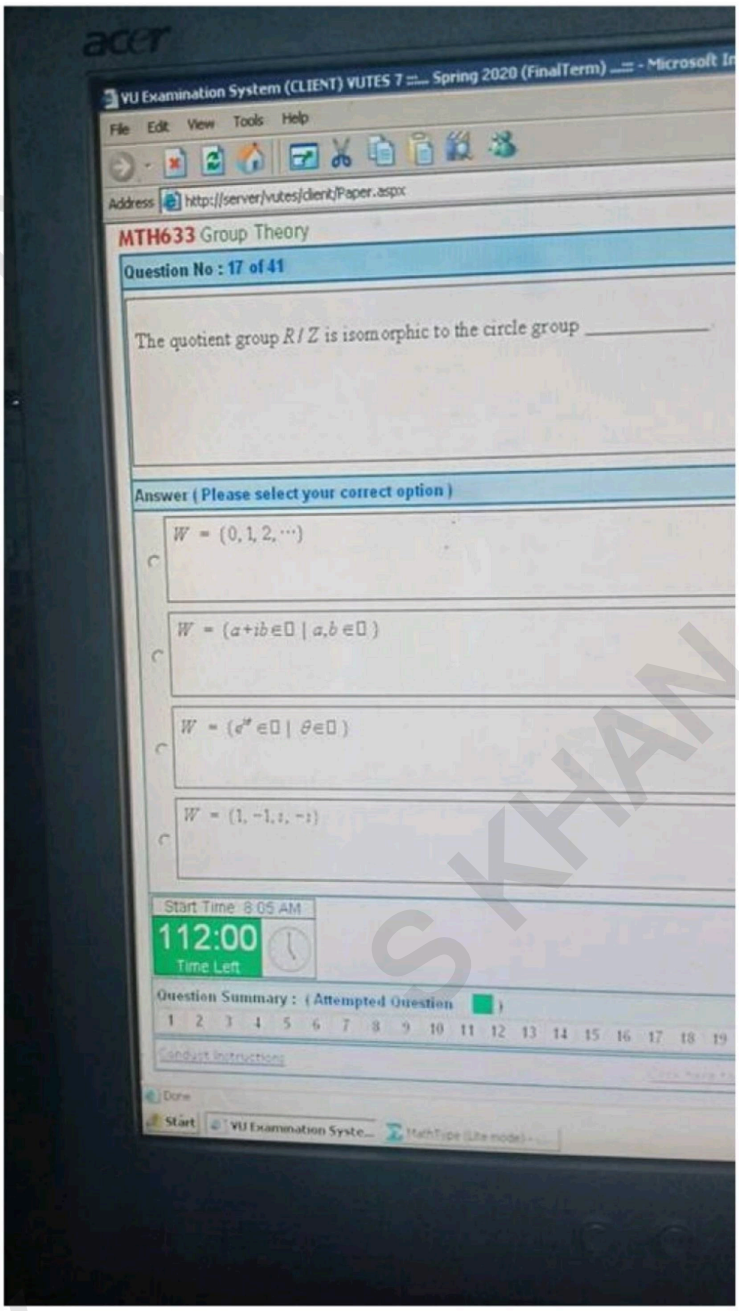

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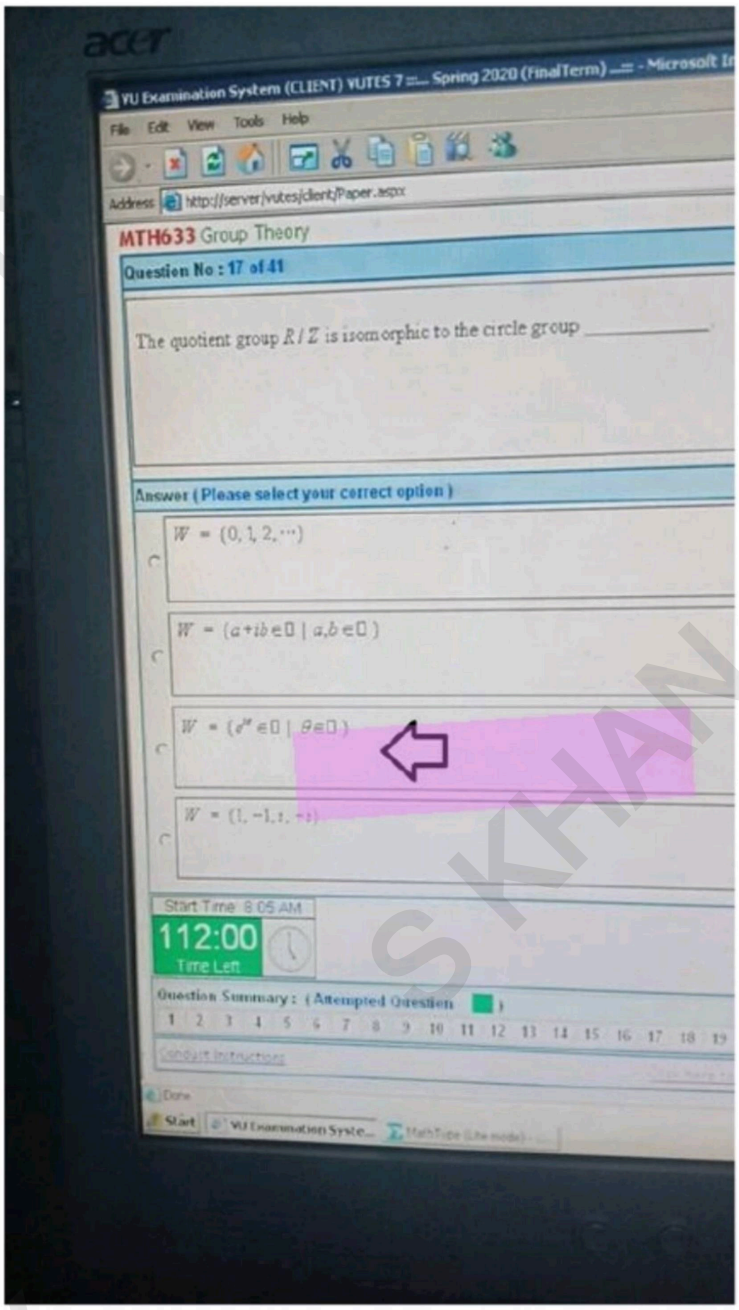

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# of Morphism Theorem

Example

Show that the quotient group $\mathbb{R}/\mathbb{Z}$ is isomorphic to the circle group

$W = \{e^{i\theta} \in \mathbb{C} \mid \theta \in \mathbb{R}\}$.

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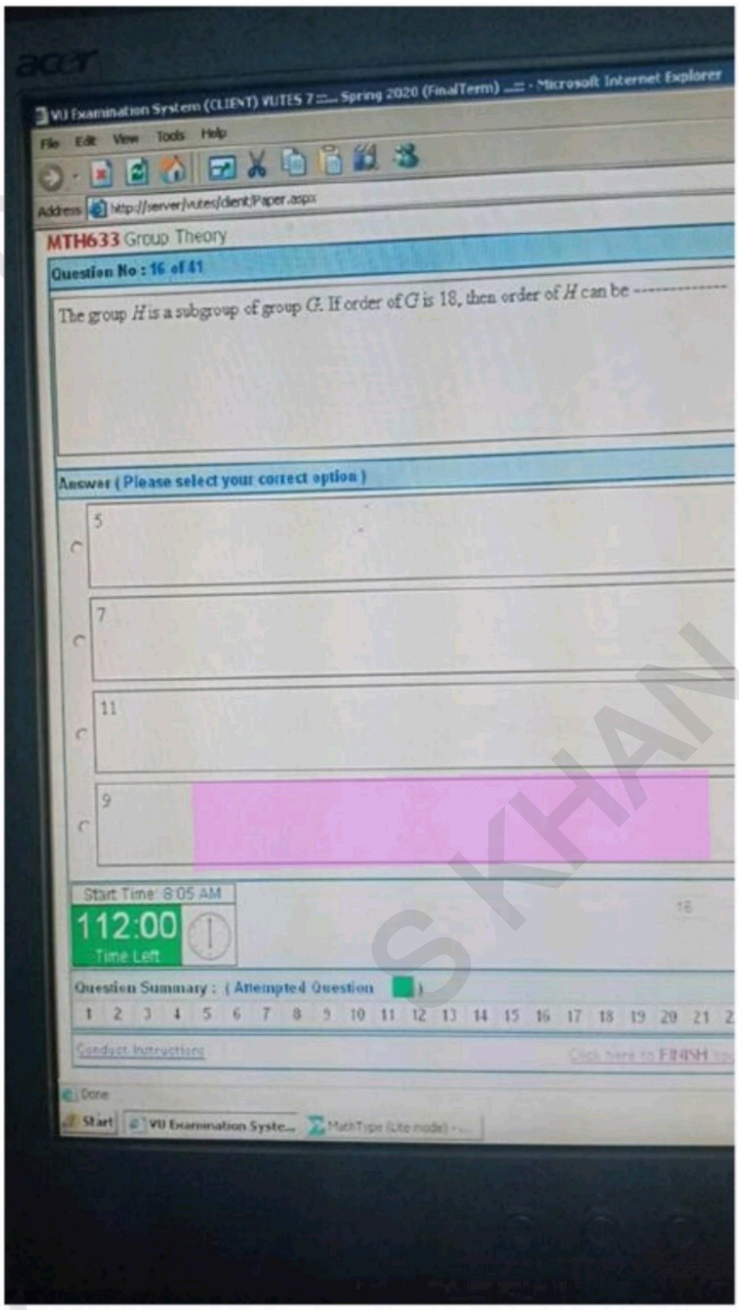

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up Computations

The theorem of Lagrange states if H is a subgroup of a finite group G, then the order of H divides the order of G.

We show that it is false that if d divides the order of G, then there must exist a subgroup H of G having order d.

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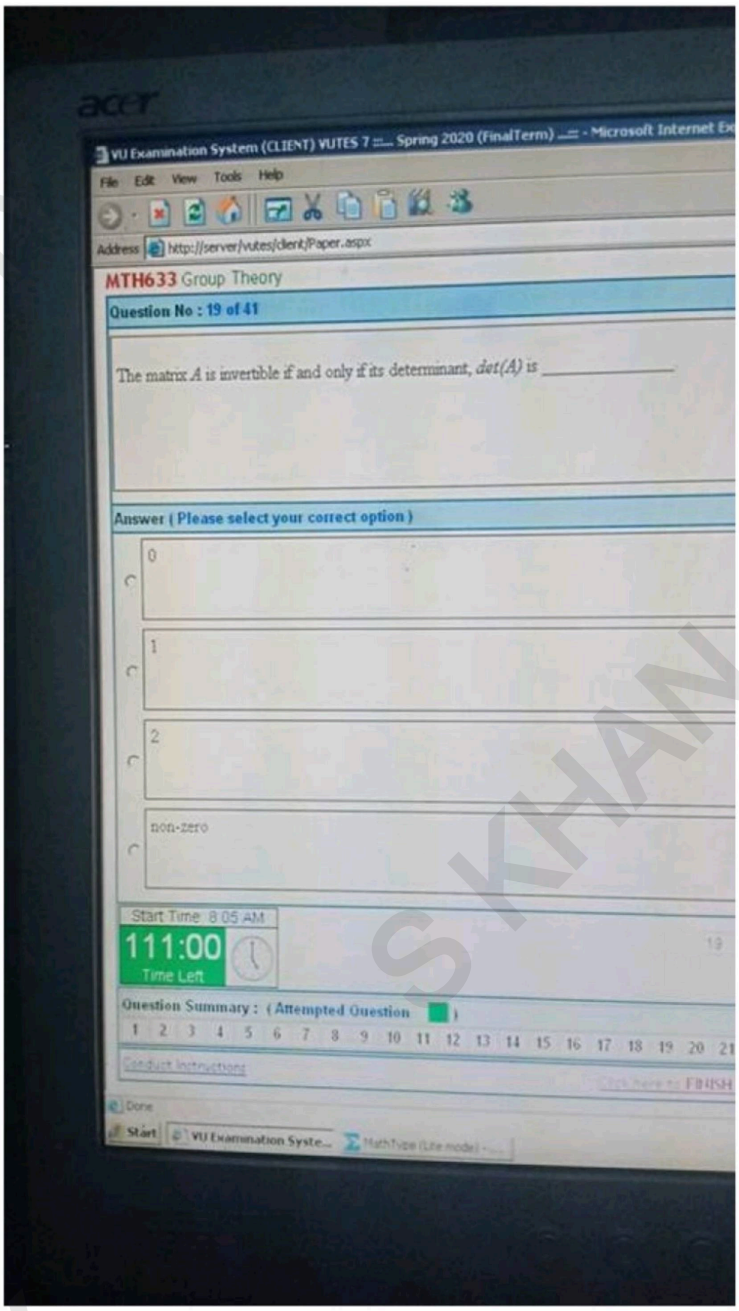

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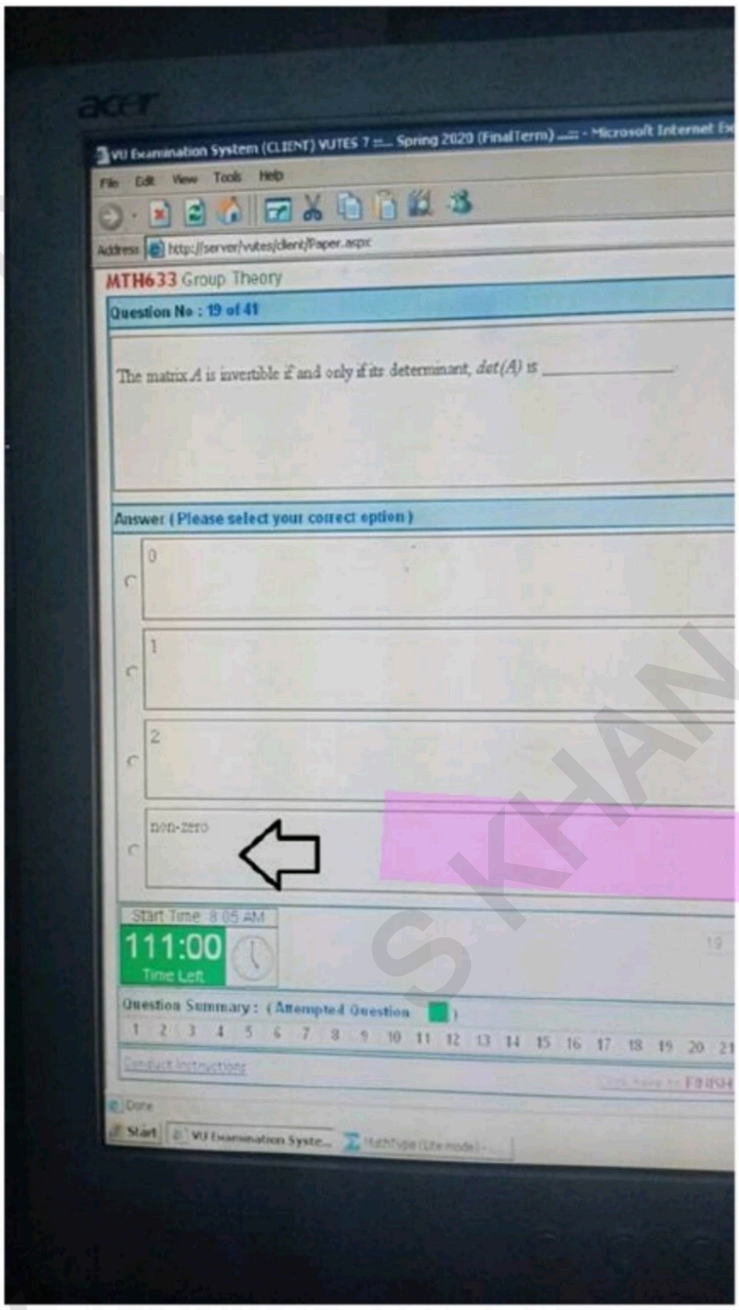

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Example

Let GL(n, $\mathbb{R}$) be the multiplicative group of all invertible n x n matrices. Recall that a matrix A is invertible if and only if its determinant, det(A), is nonzero.

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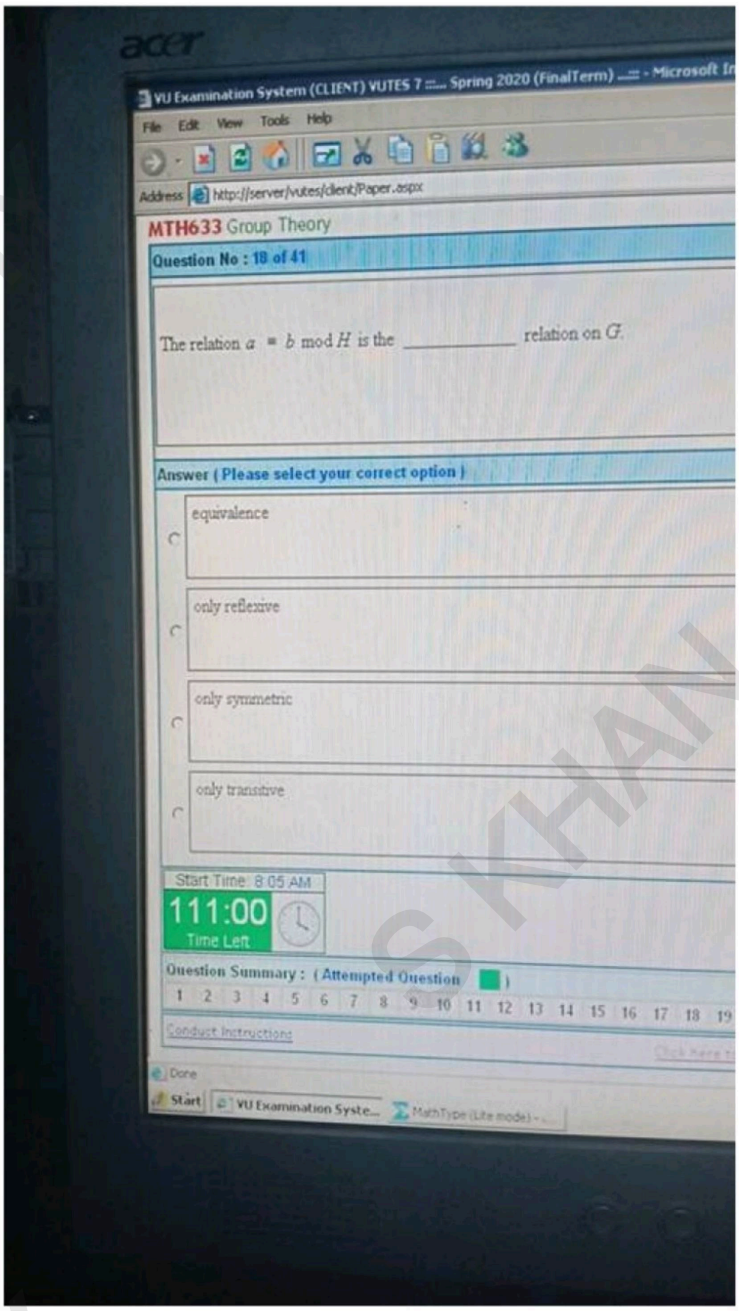

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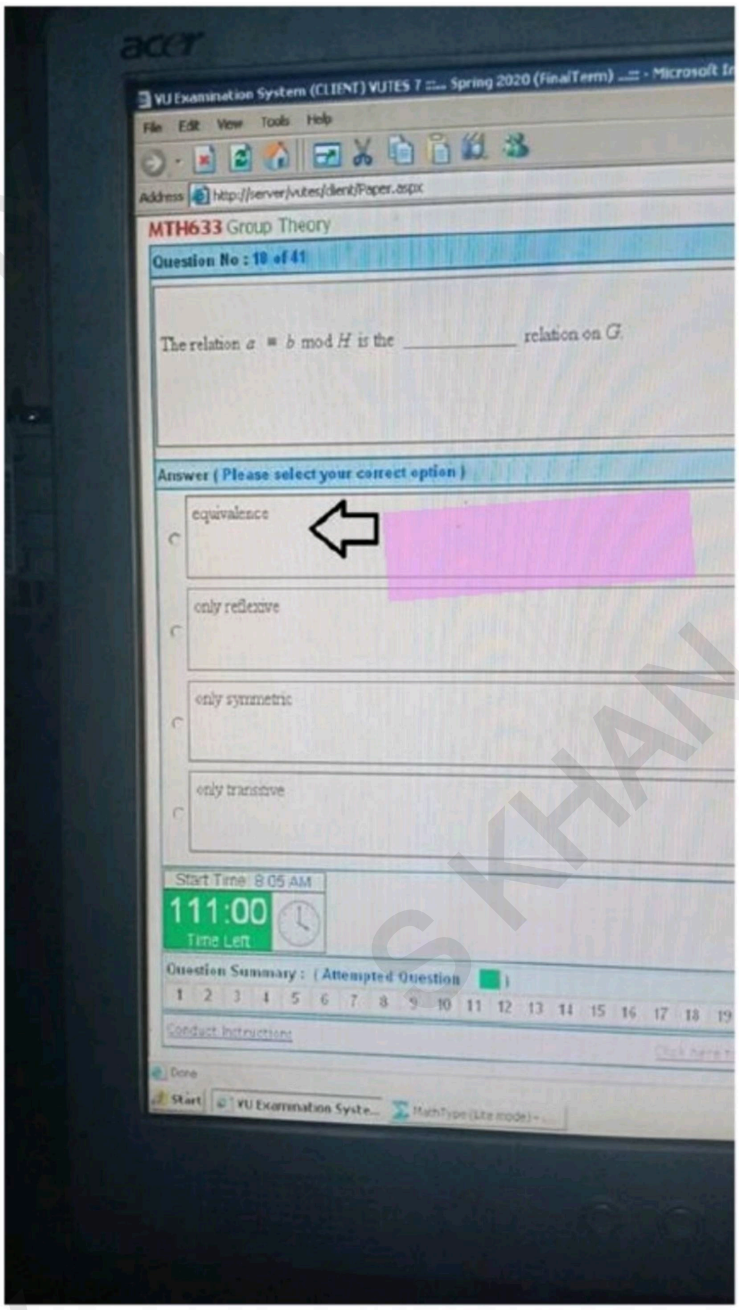

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Normal Subgroups

Normal Subgrops

Let G be a group with subgroup H. The *right cosets* of H in G are equivalence classes under the relation $a \equiv b \bmod H$, defined by $ab^{-1} \in H$. We can also define the relation L on G so that a L b if and only if $b^{-1}a \in H$. This relation, L, is an equivalence relation, and the equivalence class containing a is the *left coset* $aH = \{ah | h \in H\}$. As the following example shows, the left coset of an element does not necessarily equal the right coset.

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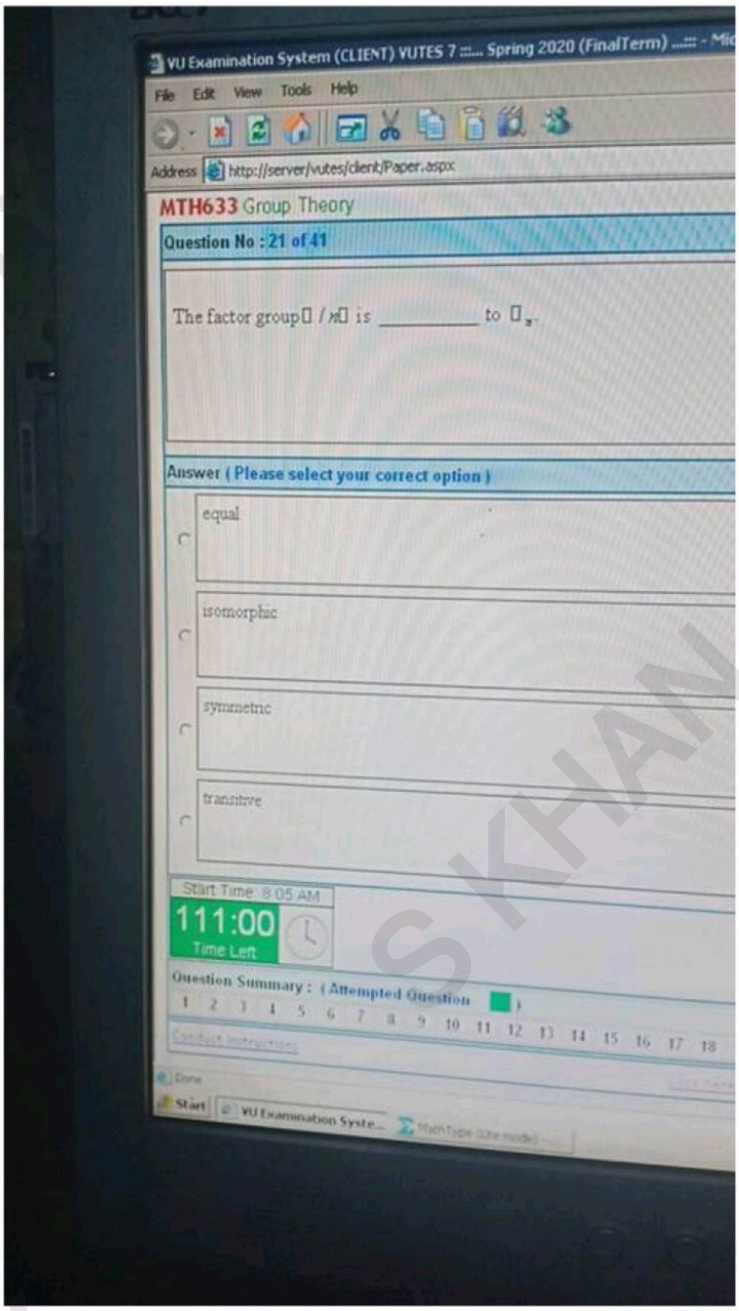

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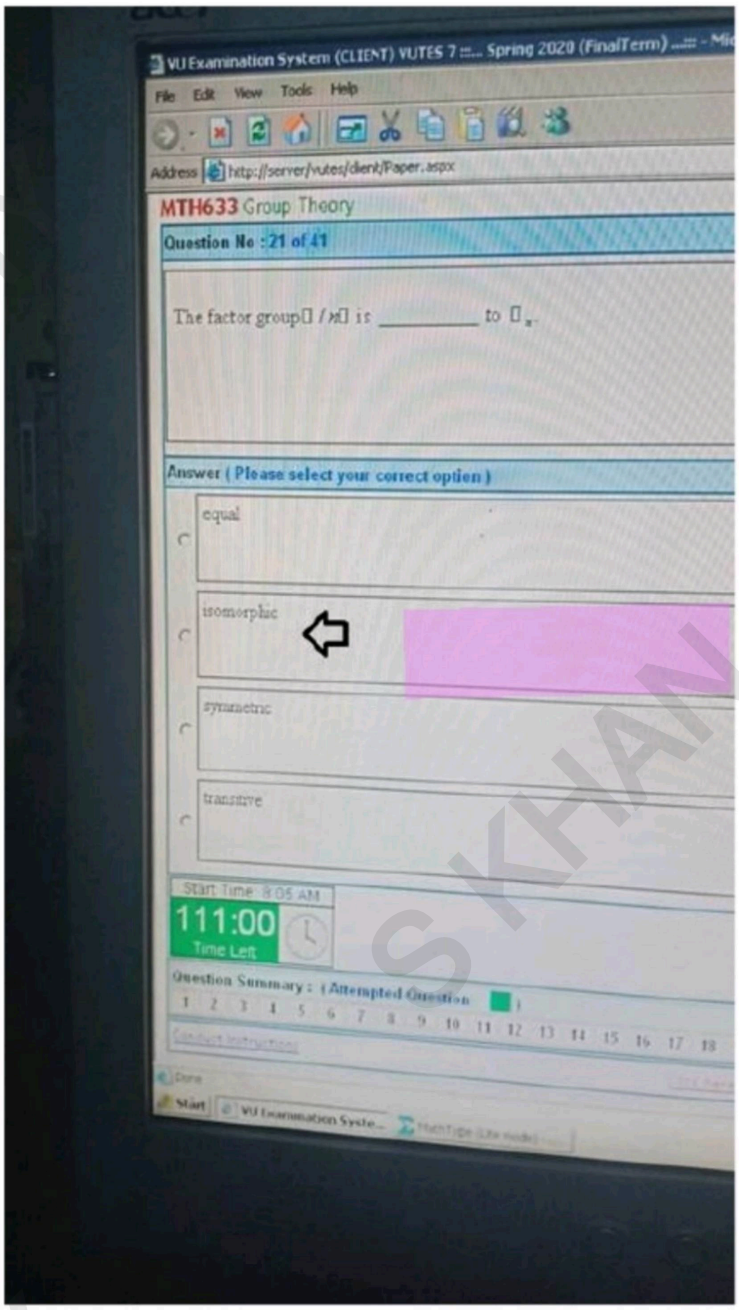

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Factor Groups from Homomorphisms

Example

Consider the map y: $\mathbb{Z} \to \mathbb{Z}_n$, where y(m) is the remainder when m is divided by n in accordance with the division algorithm. We know that *y* is a homomorphism. Of course, Ker(y) = $n\mathbb{Z}$. By above Theorem, we see that the factor group $\mathbb{Z}/n\mathbb{Z}$ is isomorphic to $\mathbb{Z}_n$. The cosets of $n\mathbb{Z}$ are the residue classes modulo n.

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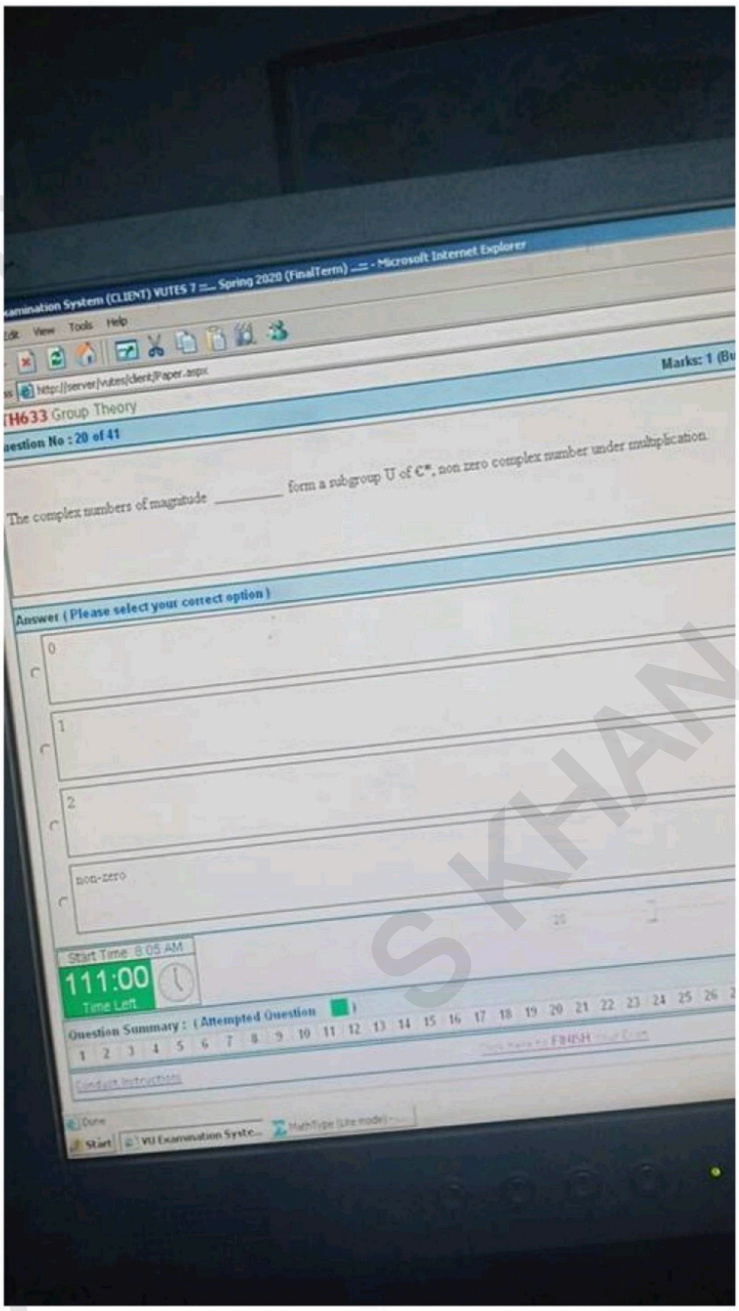

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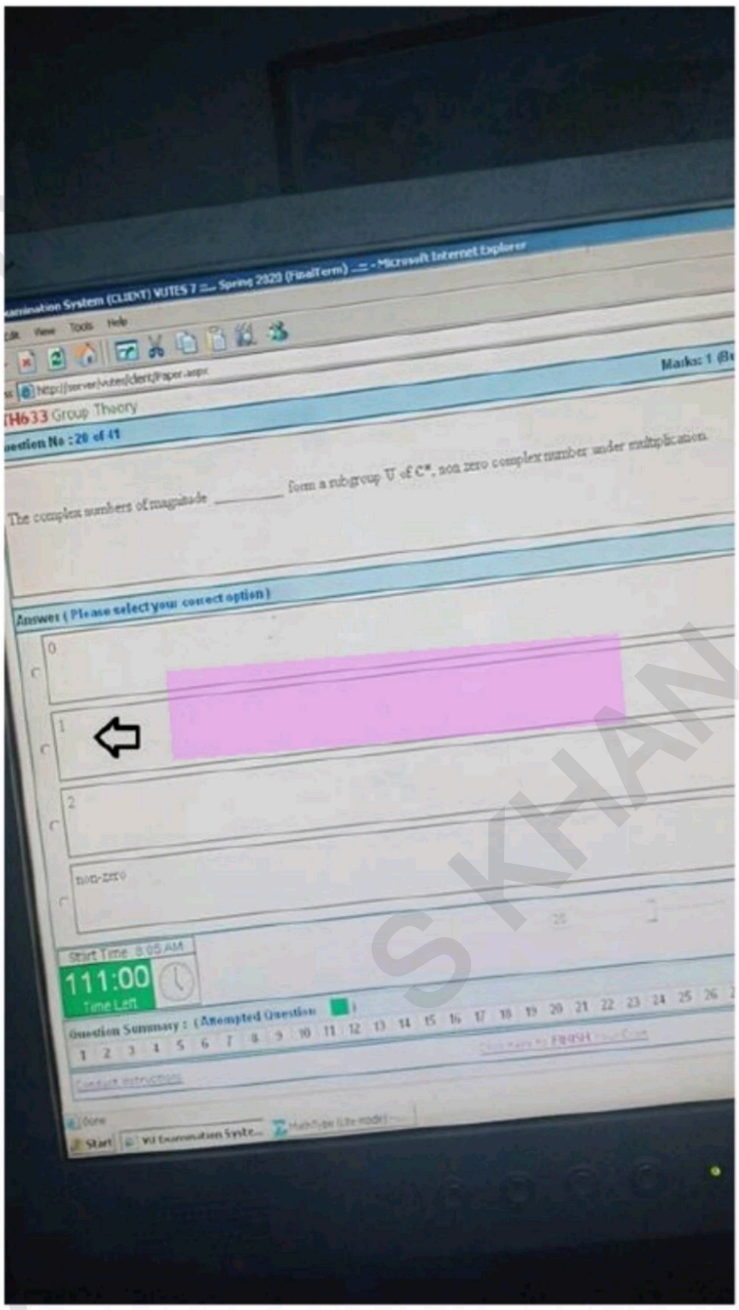

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Kernel of a Homomorphism

Since {1} is a subgroup of $\mathbb{R}^+$, the complex numbers of magnitude 1 form a subgroup U of $\mathbb{C}^*$. Recall that the complex numbers can be viewed as filling the coordinate plane, and that the magnitude of a complex number is its distance from the origin. Consequently, the cosets of U are circles with center at the origin. Each circle is collapsed by this homomorphism onto its point of intersection with the positive real axis.

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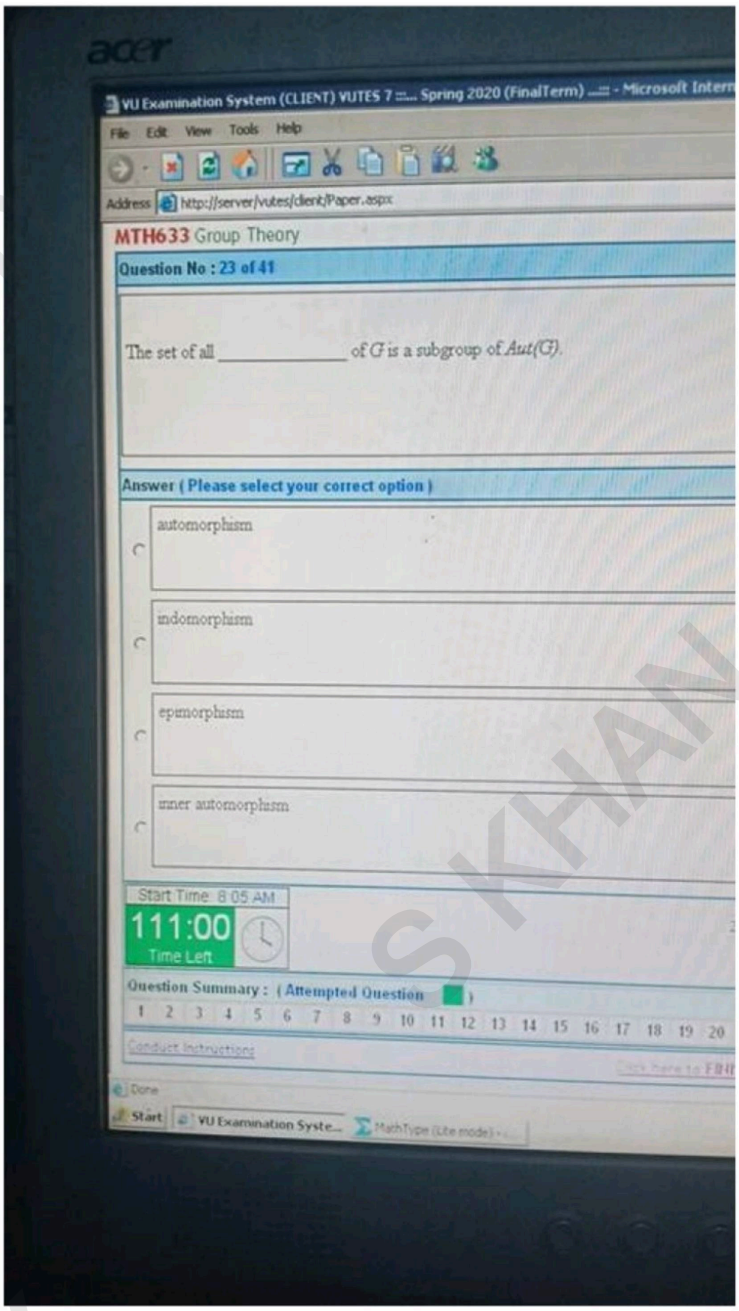

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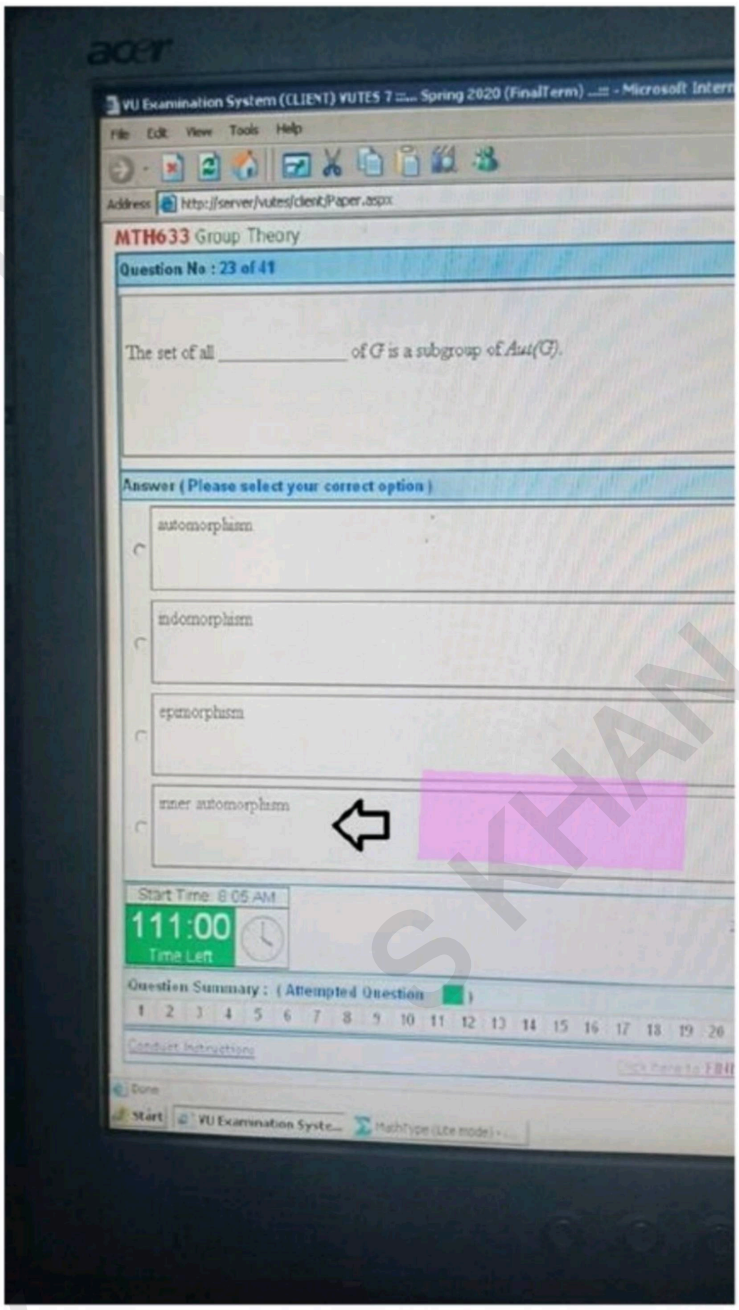

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Inner Automorphisms

Lemma

The set of all inner automorphisms of G is a subgroup of Aut(G).

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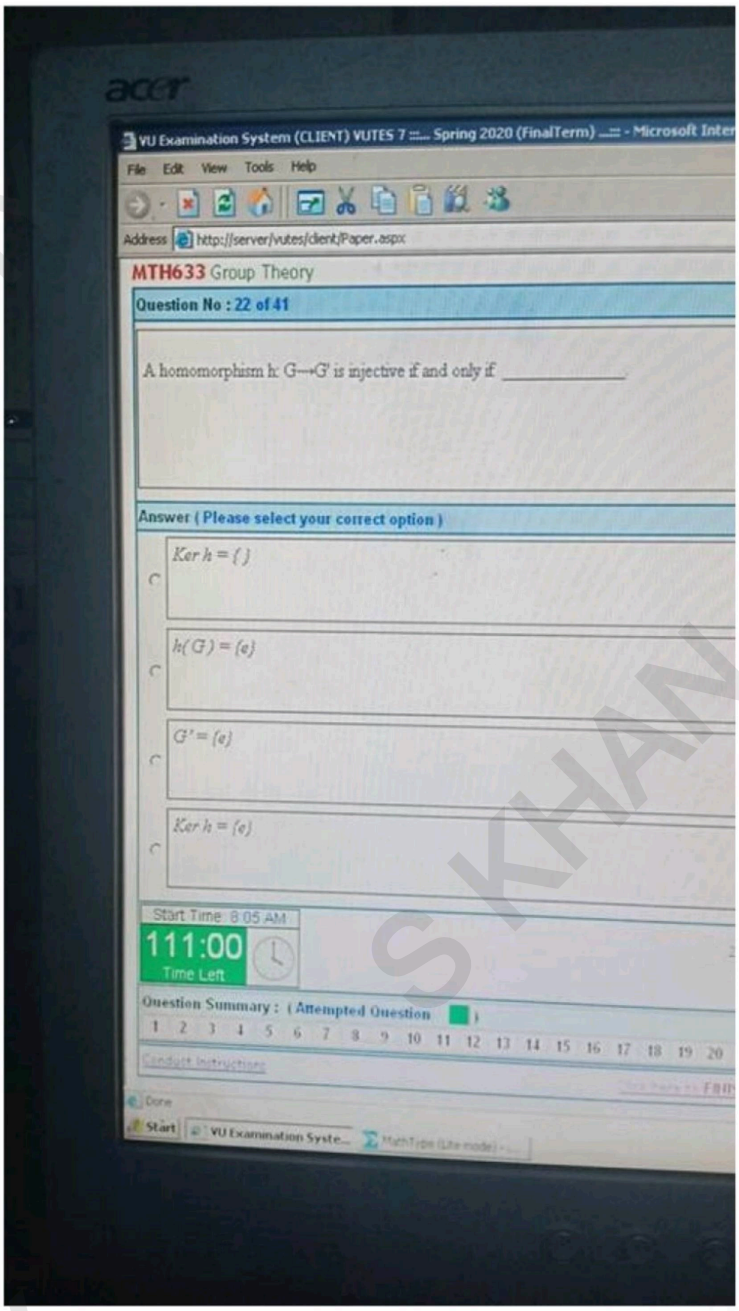

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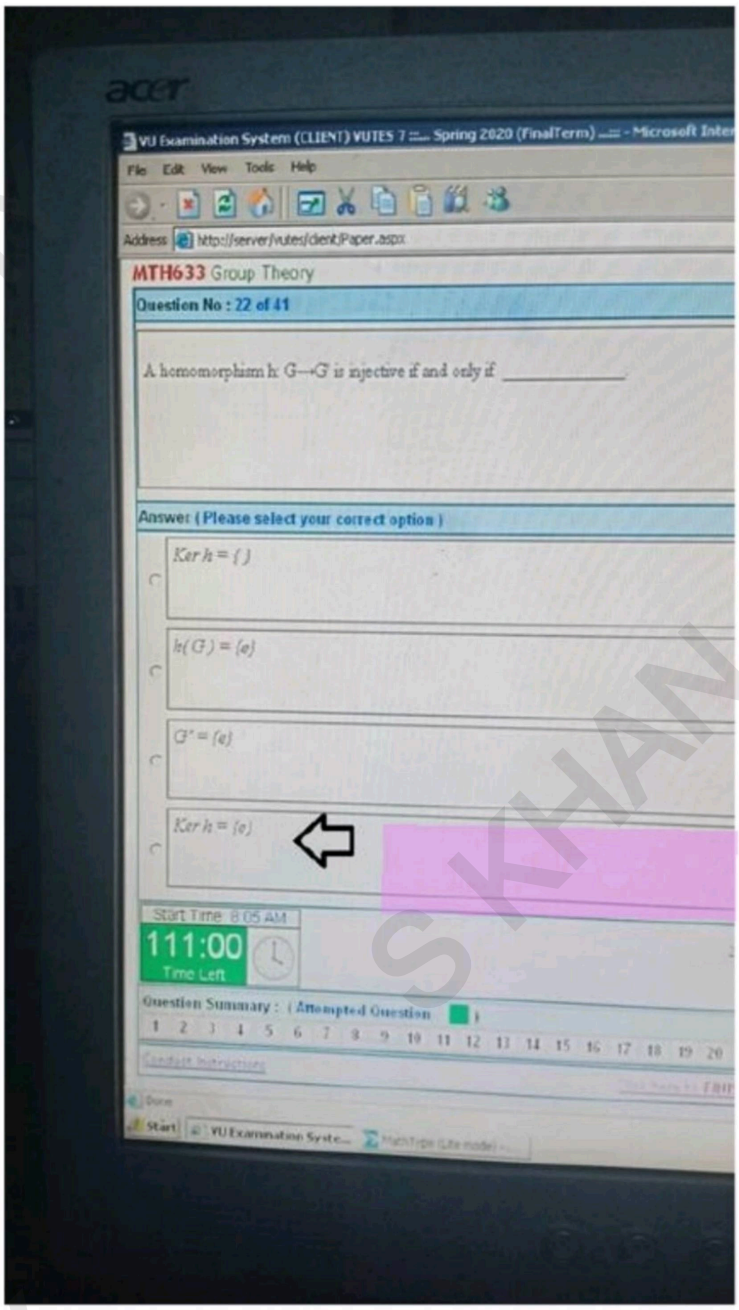

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Properties of Homomorphisms

Corollary

Let ϕ: G → G' be a group morphism. Then, ϕ is injective if and only if Ker ϕ = {e}.

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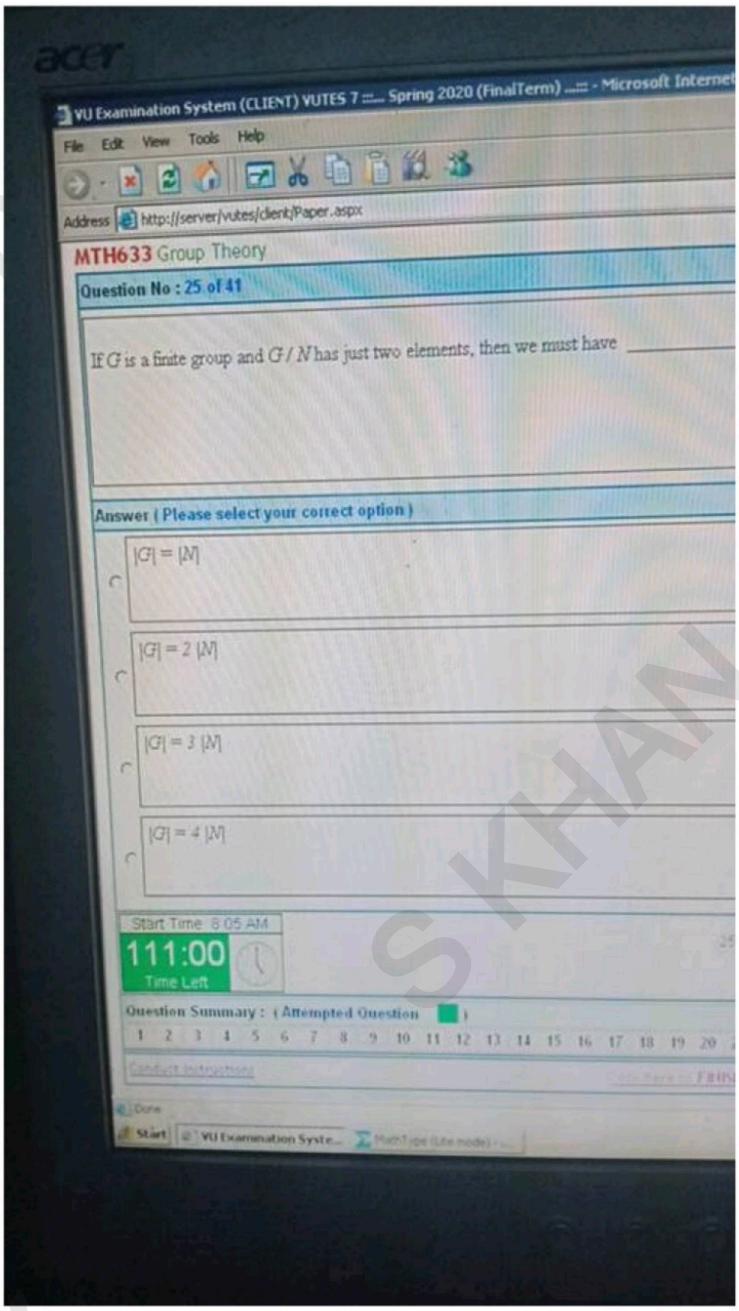

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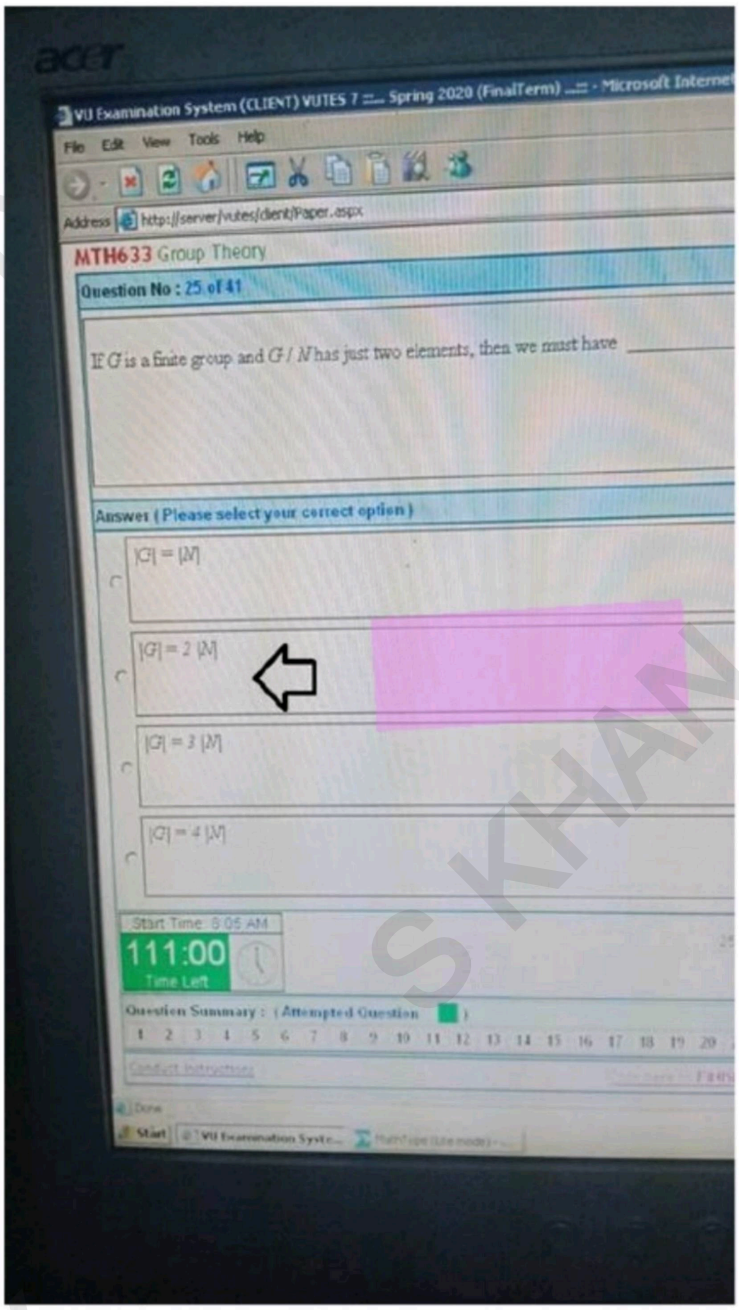

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or Group Computations

In next module, we give example showing that even when G/N has order 2, we may be able to deduce some useful results.

If G is a finite group and G/N has just two elements, then we must have $|G|=2|N|$.

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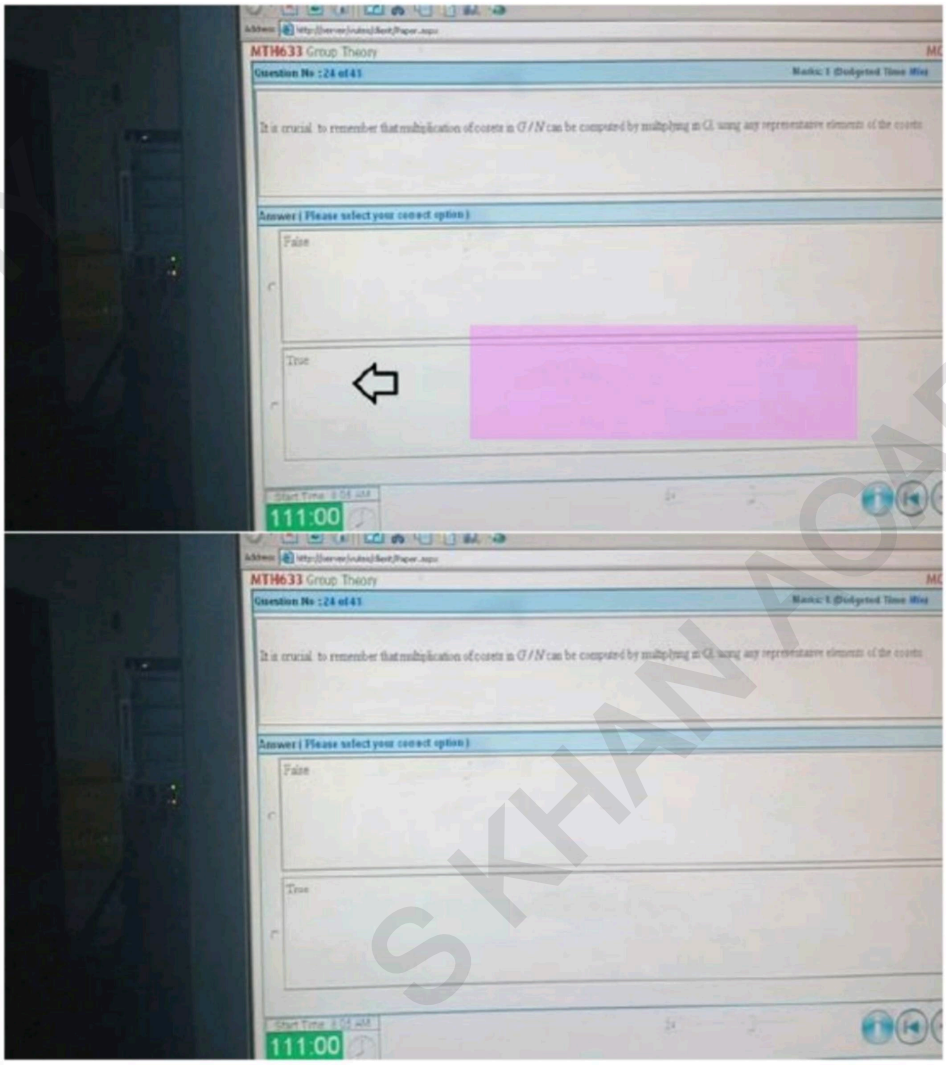

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Factor Group Computations

Recall that $y: G \to G/N$ defined by $y(a)=aN$ for $a \in G$ is a homomorphism of G onto G / N. We can view the "line" G / N at the bottom of the figure as obtained by collapsing to a point each coset of N in another copy of G. Each point of G / N thus corresponds to a whole vertical line segment in the shaded portion, representing a coset of N in G. It is crucial to remember that multiplication of cosets in G / N can be computed by multiplying in G, using any representative elements of the cosets.

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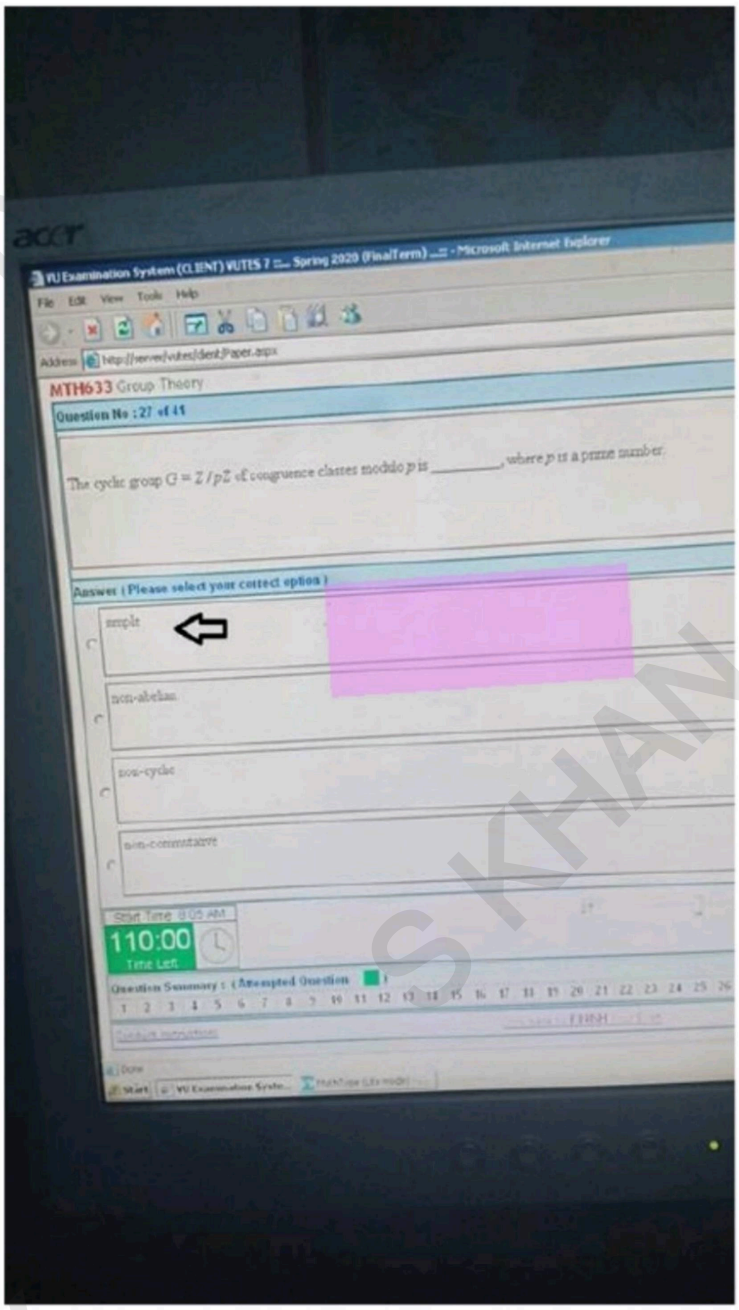

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Simple Groups

Example

The cyclic group $G=\mathbb{Z}/p\mathbb{Z}$ of congruence classes modulo p is simple, where p is a prime number.

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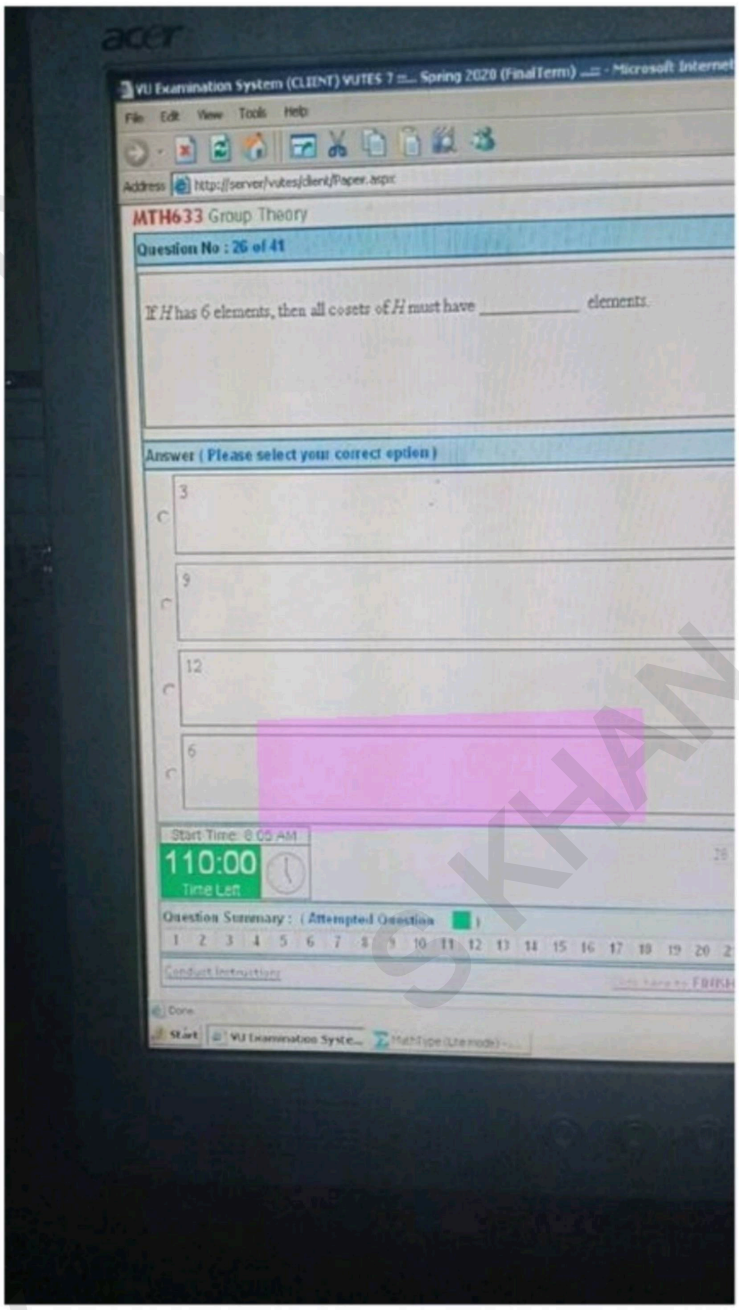

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Factor Group Computations

Example

Let us compute the factor group $(\mathbb{Z}_4 x \mathbb{Z}_6)/\langle(0, 1)\rangle$. Here $\langle(0, 1)\rangle$ is the cyclic subgroup H of $\mathbb{Z}_4 x \mathbb{Z}_6$ generated by (0, 1). Thus

H = {(0, 0), (0, 1), (0. 2), (0, 3), (0, 4), (0, 5)}.

Since $\mathbb{Z}_4 x \mathbb{Z}_6$ has 24 elements and H has 6 elements, all cosets of H must have 6 elements, and $(\mathbb{Z}_4 x\ \mathbb{Z}_6)/H$ must have order 4. Since $\mathbb{Z}_4 x \mathbb{Z}_6$ is abelian, so is $(\mathbb{Z}_4 x\ \mathbb{Z}_6)/H$. Remember, we compute in a factor group by means of representatives from the original group.

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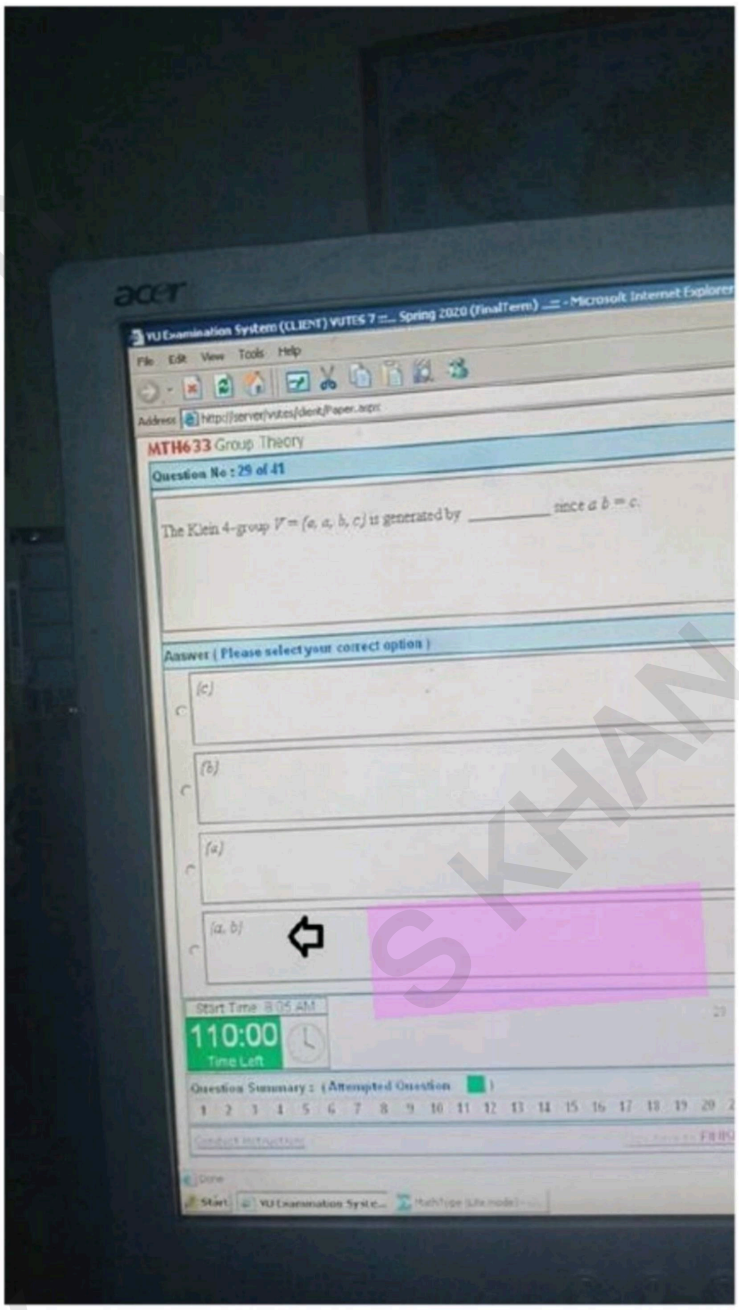

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enerating Sets

Example

The Klein 4-group V = {e, a, b, c} is generated by {a,b} since ab=c.

It is also generated by {a,c}, {b,c }, and {a,b,c}.

If a group G is generated by a subset S, then every subset of G containing S generates G.

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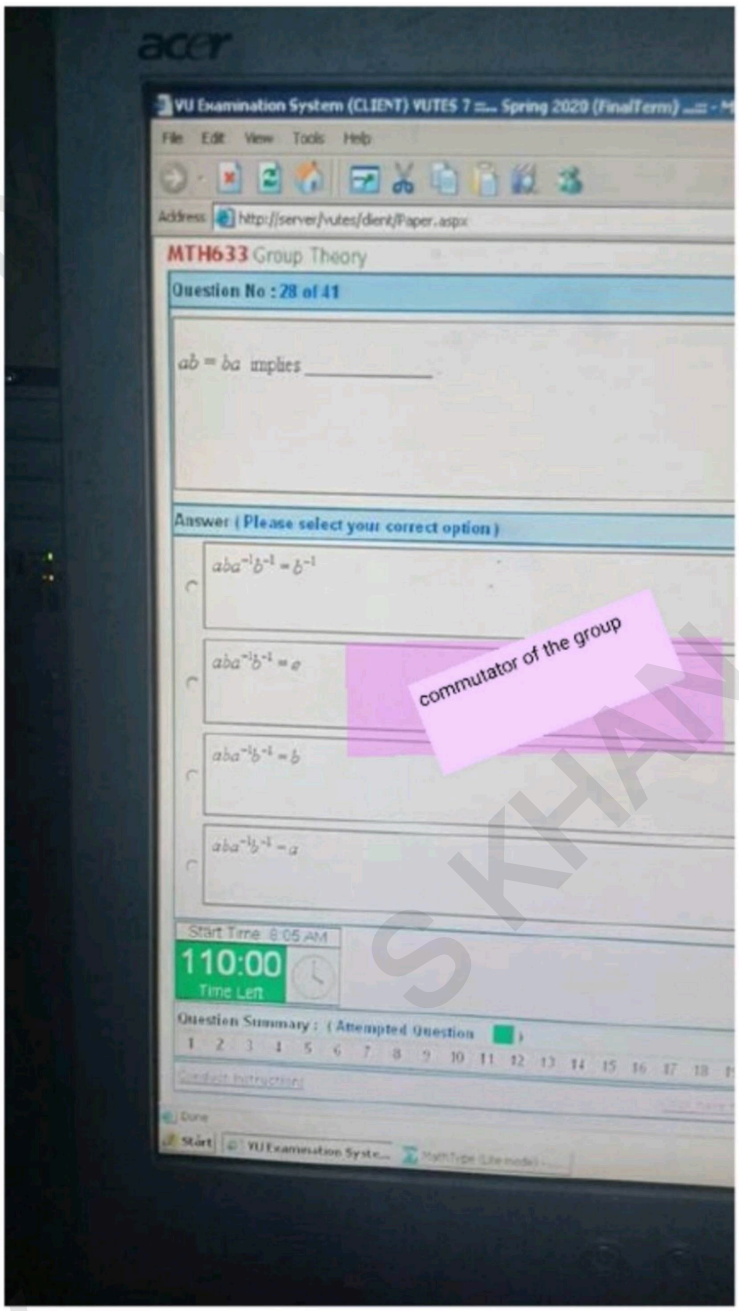

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The Commutator Subgroup

To require that ab=ba is to say that $aba^{-1}b^{-1}=e$ in our new group.

An element $aba^{-1}b^{-1}$ in a group is a **commutator of the group.**

Thus we wish to attempt to form an abelianized version of G by replacing every commutator of G by e.

We should then attempt to form the factor group of G modulo the smallest normal subgroup we can find that contains all commutators of G.

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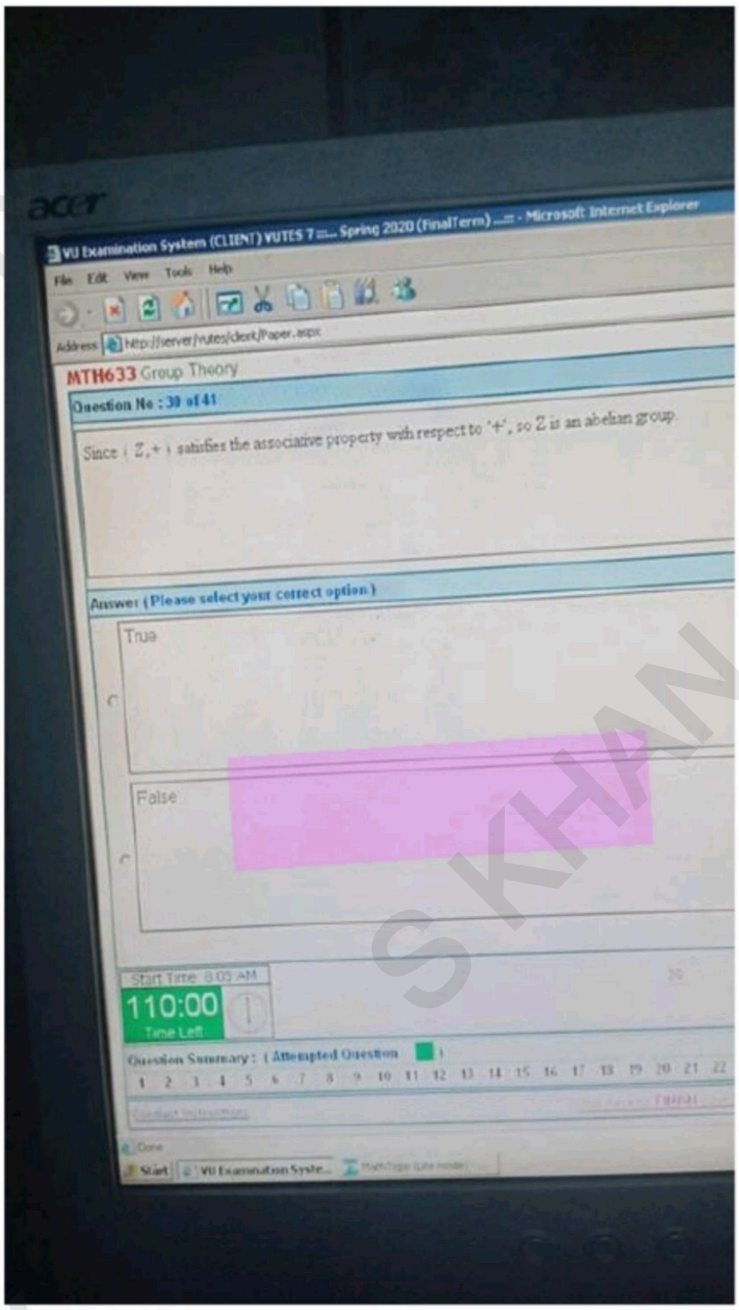

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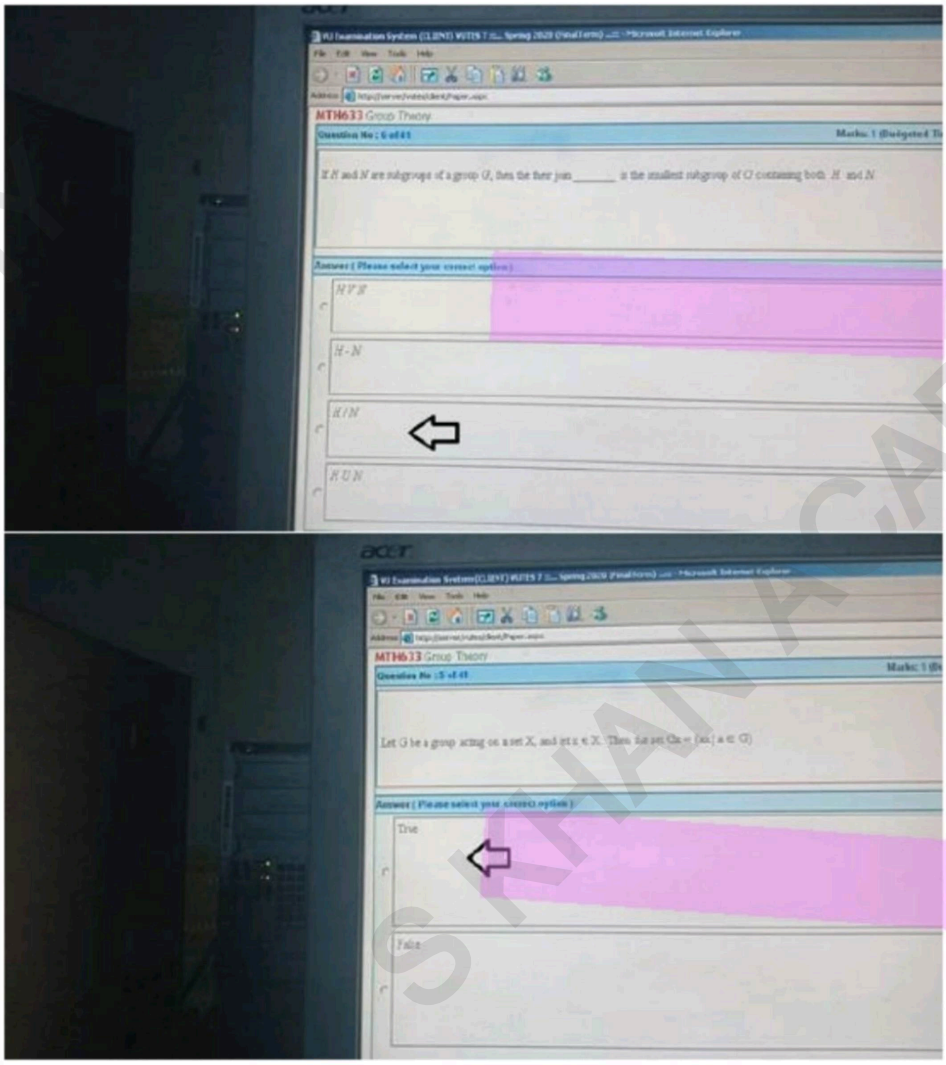

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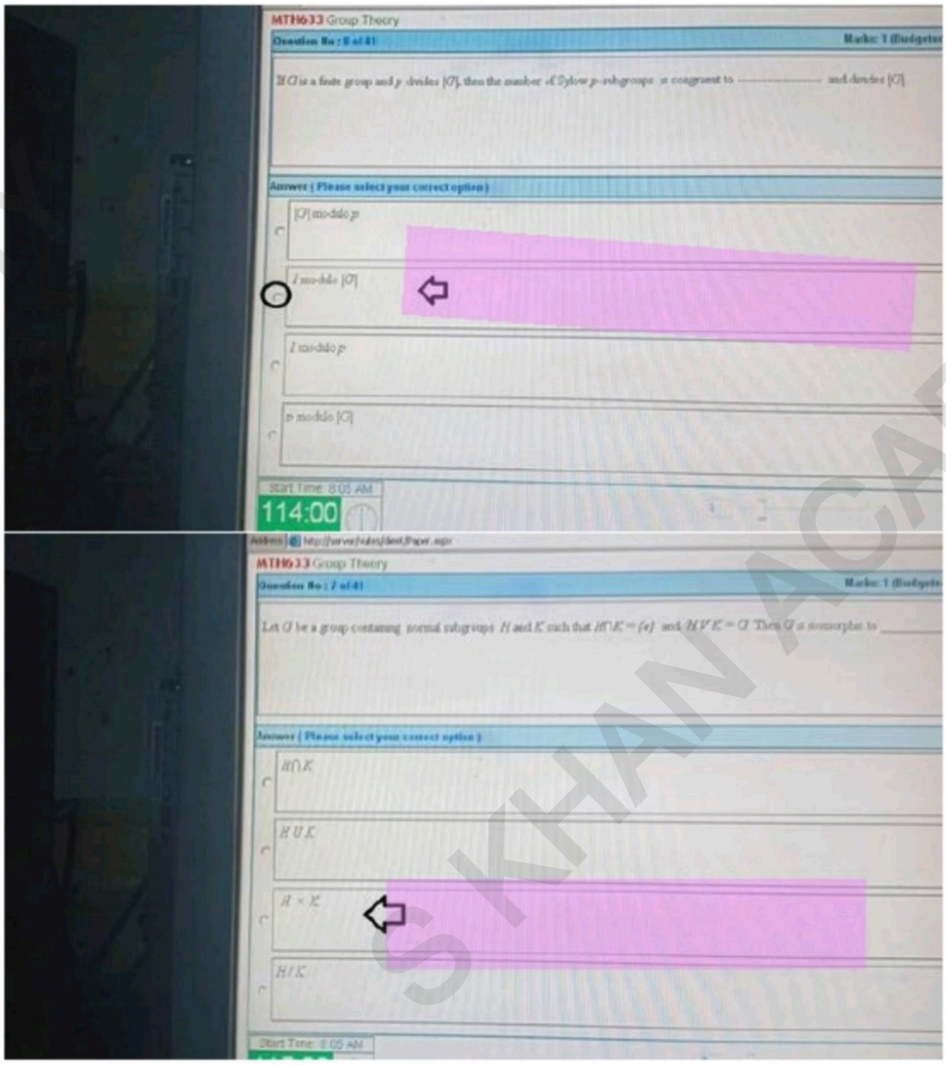

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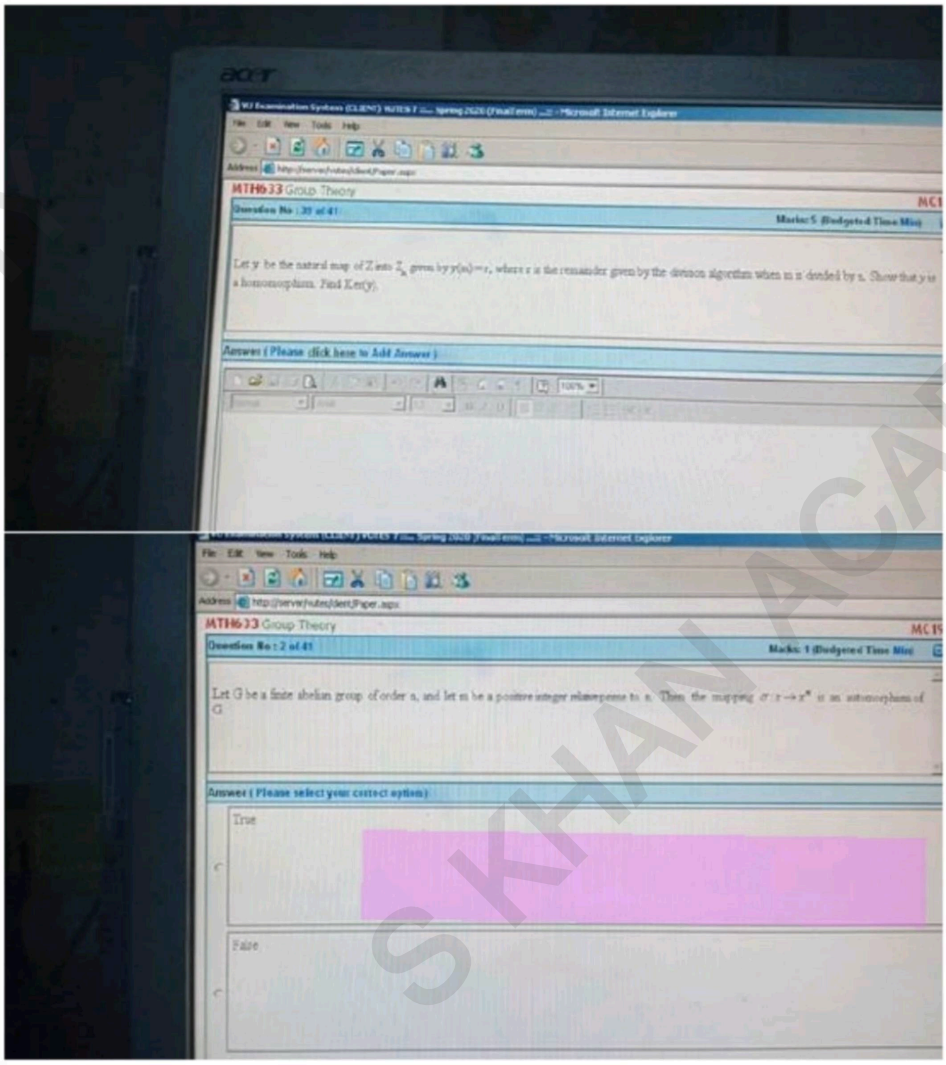

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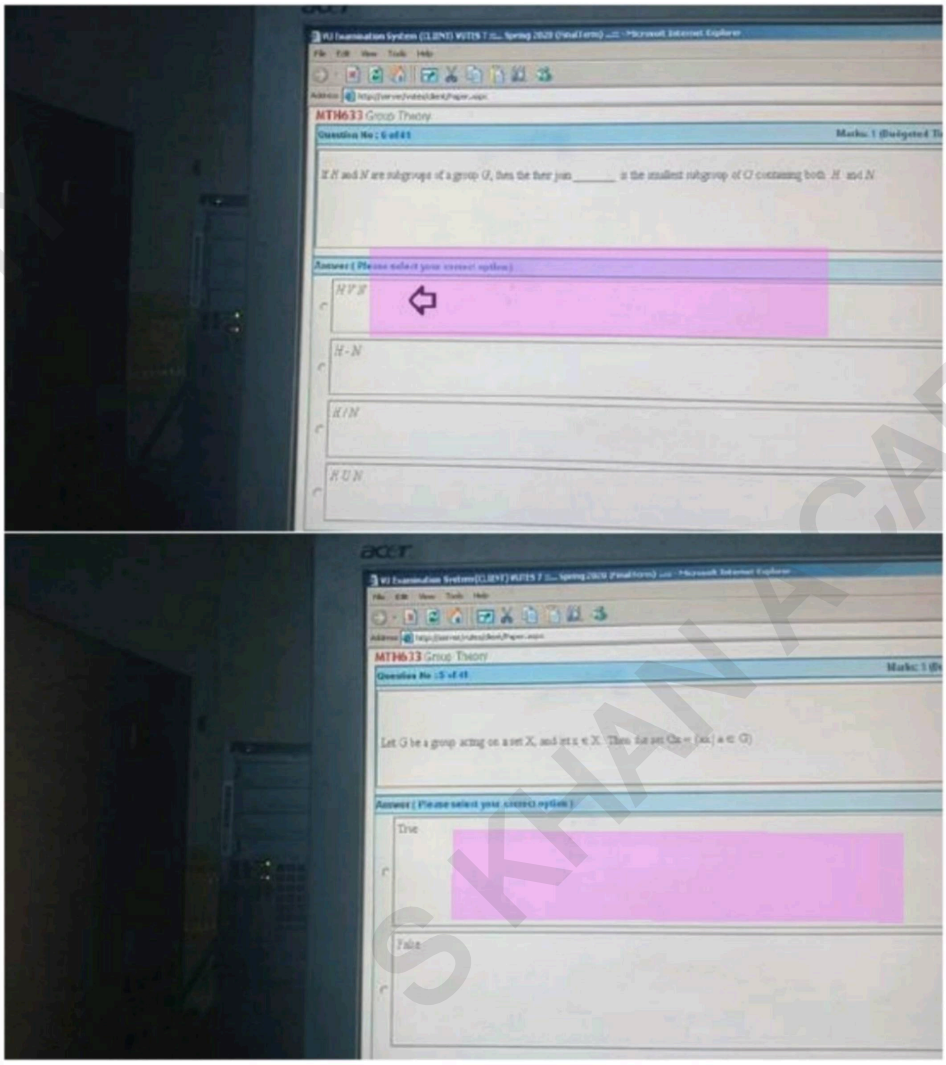

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hism Theorems

If H and N are subgroups of a group G, then we let

$HN=\{hn \mid h \in H, n \in N\}$.

We define the **join** H V N of H and N as the intersection of all subgroups of G that contain HN; thus H V N is the smallest subgroup of G containing HN.

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Definition

Let G be a group acting on a set X, and let $x \in X$. Then the set

$G_x = \{g \in G \mid gx = x\}$,

which can be shown to be a subgroup, is called the stabilizer (or isotropy) group of x in G.

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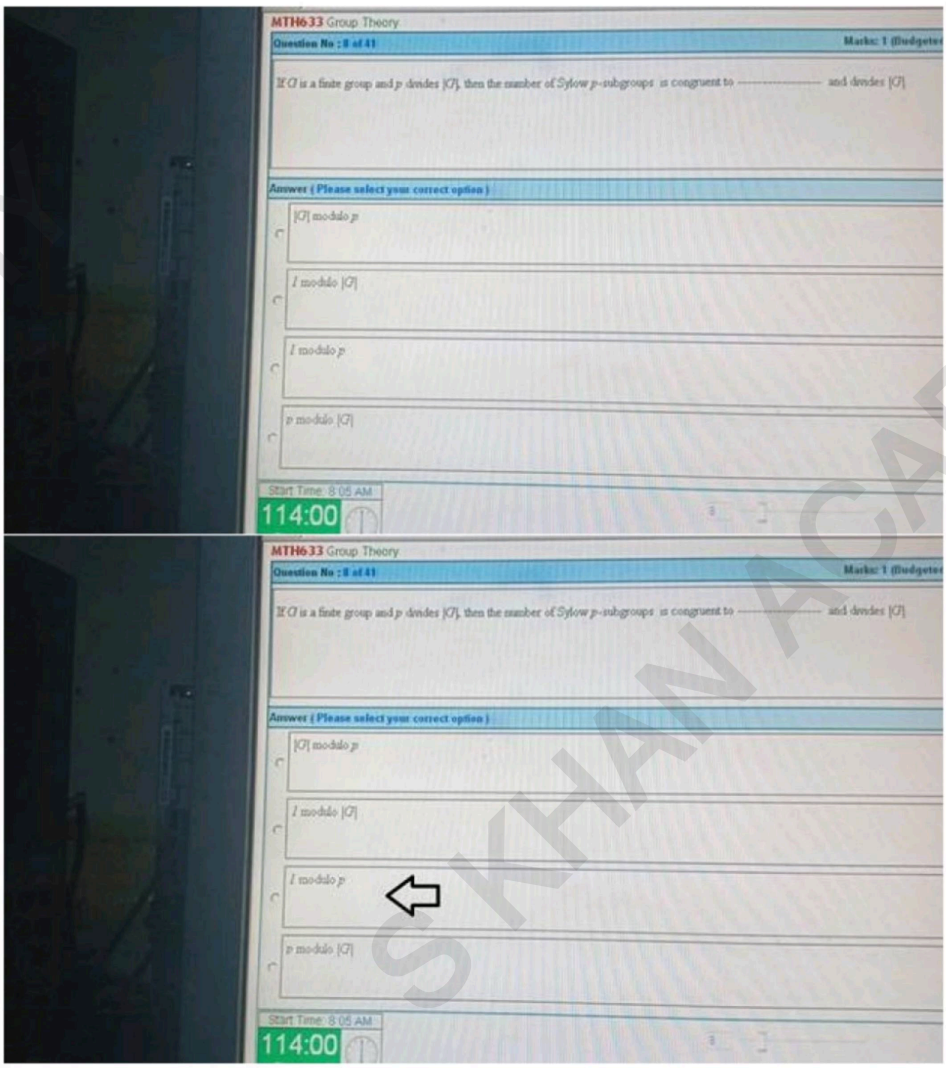

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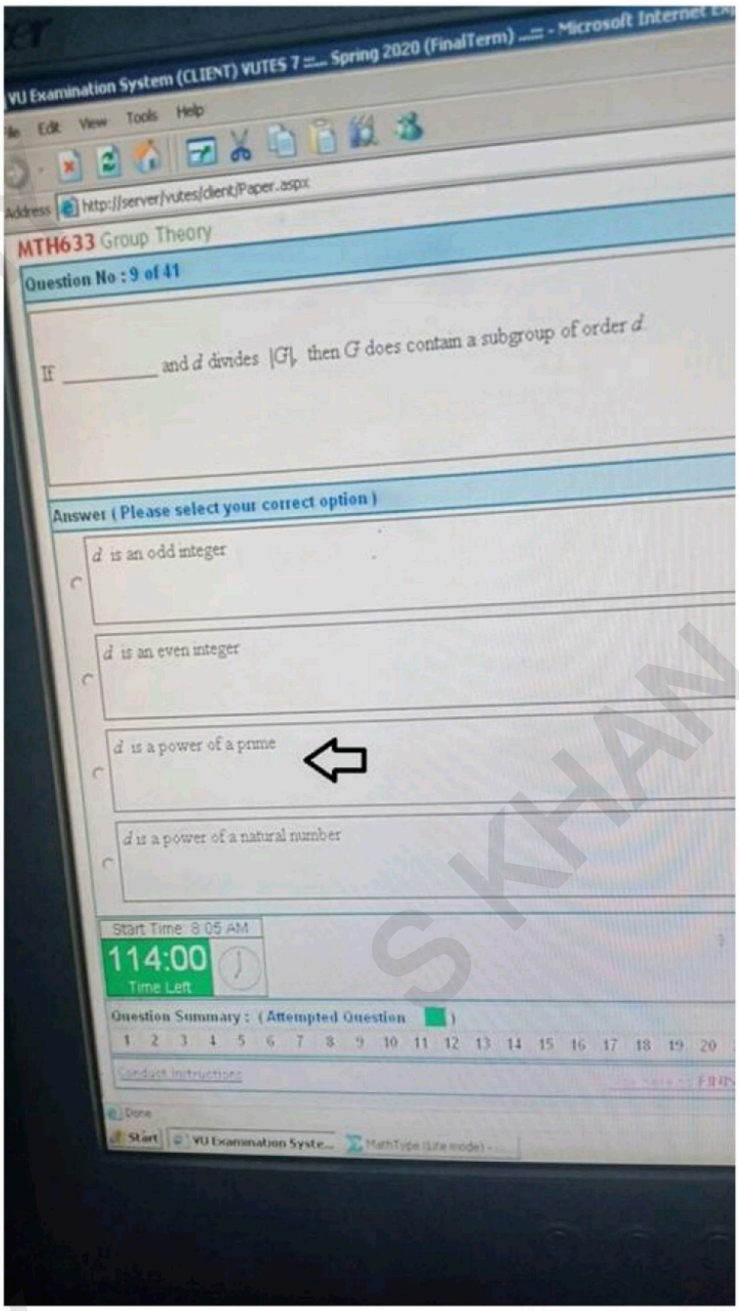

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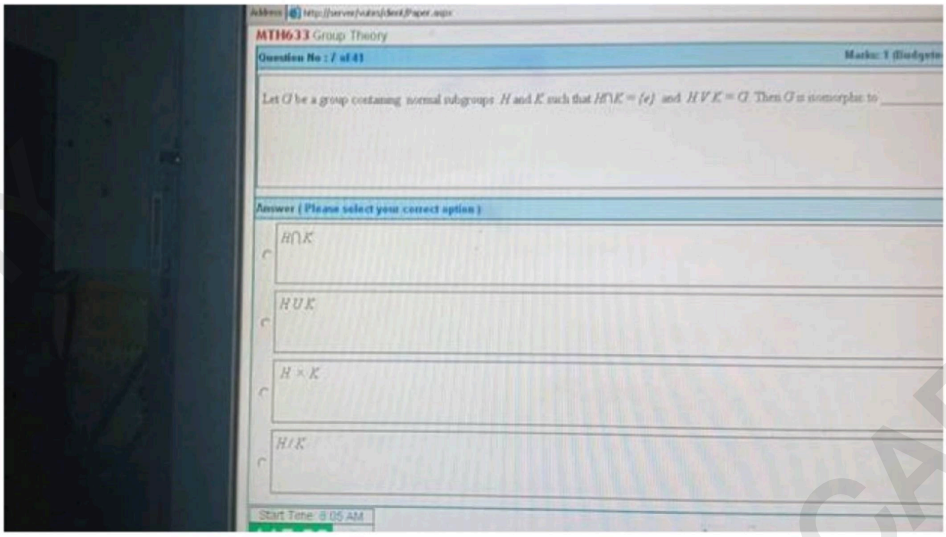

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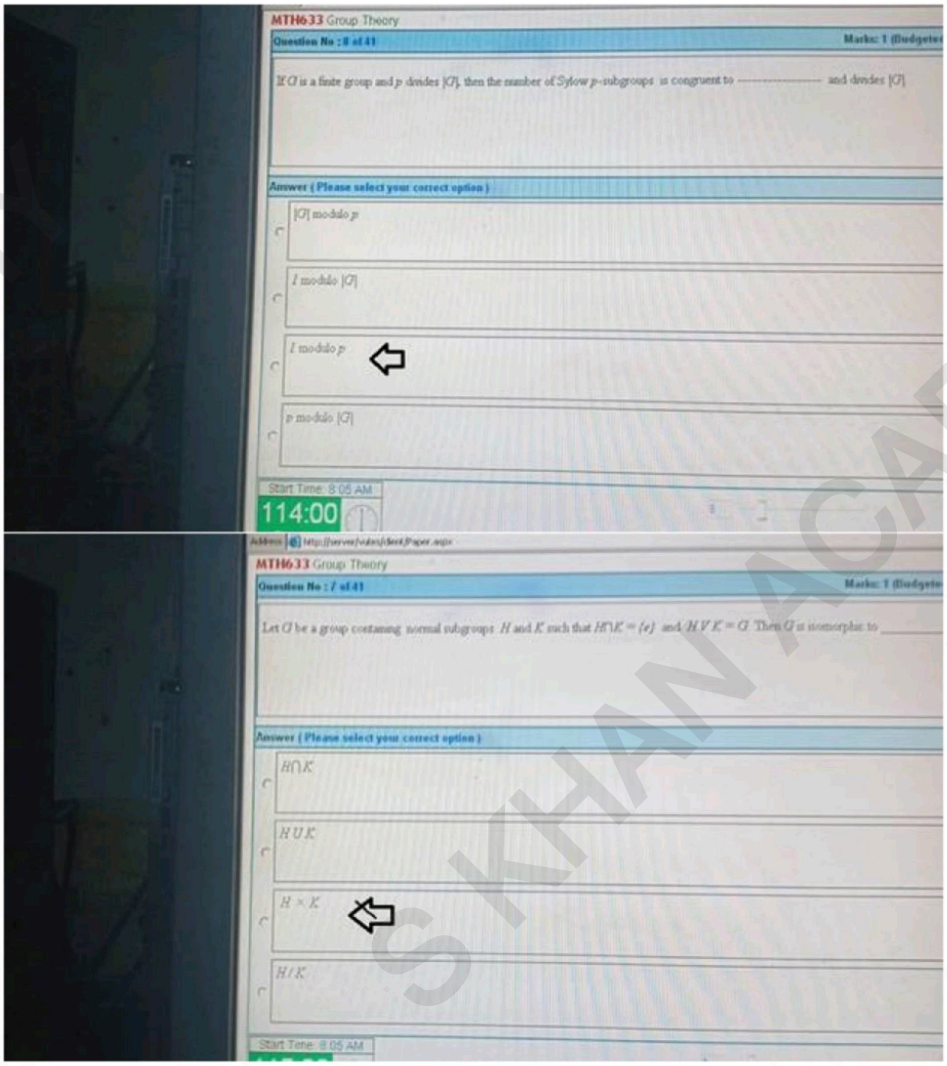

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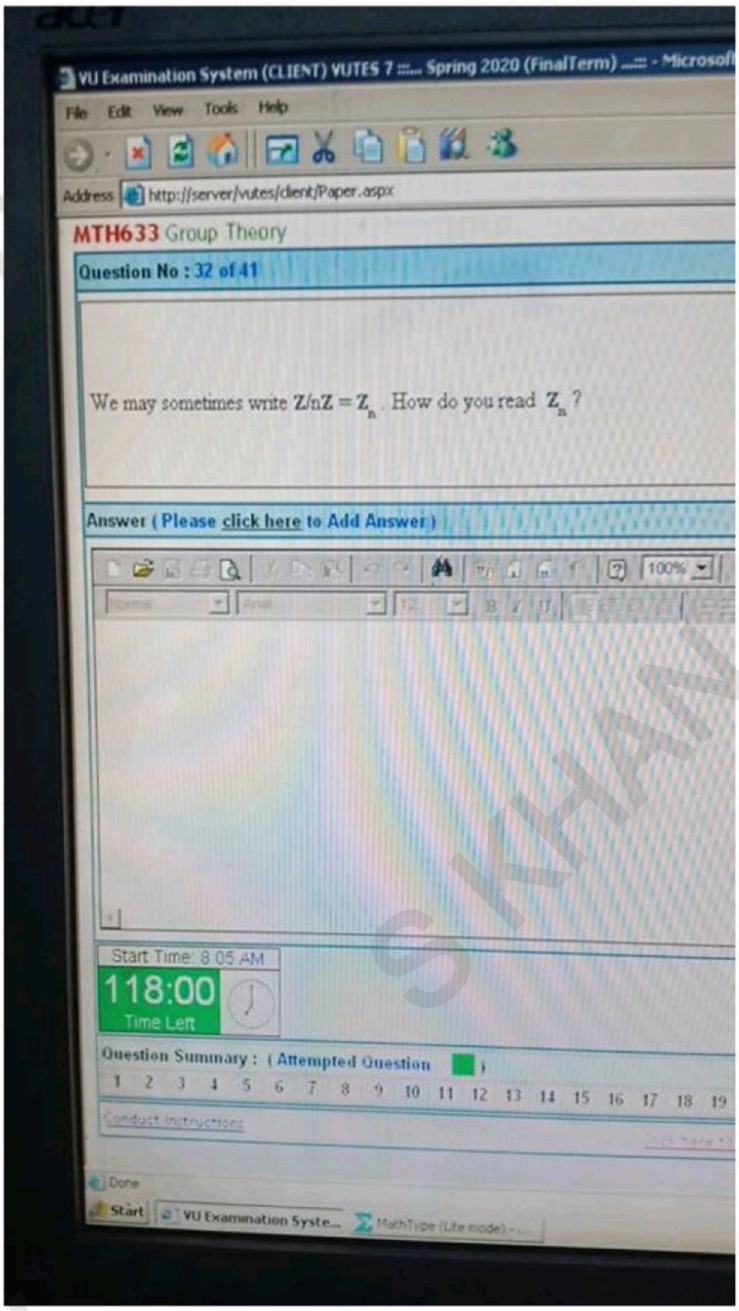

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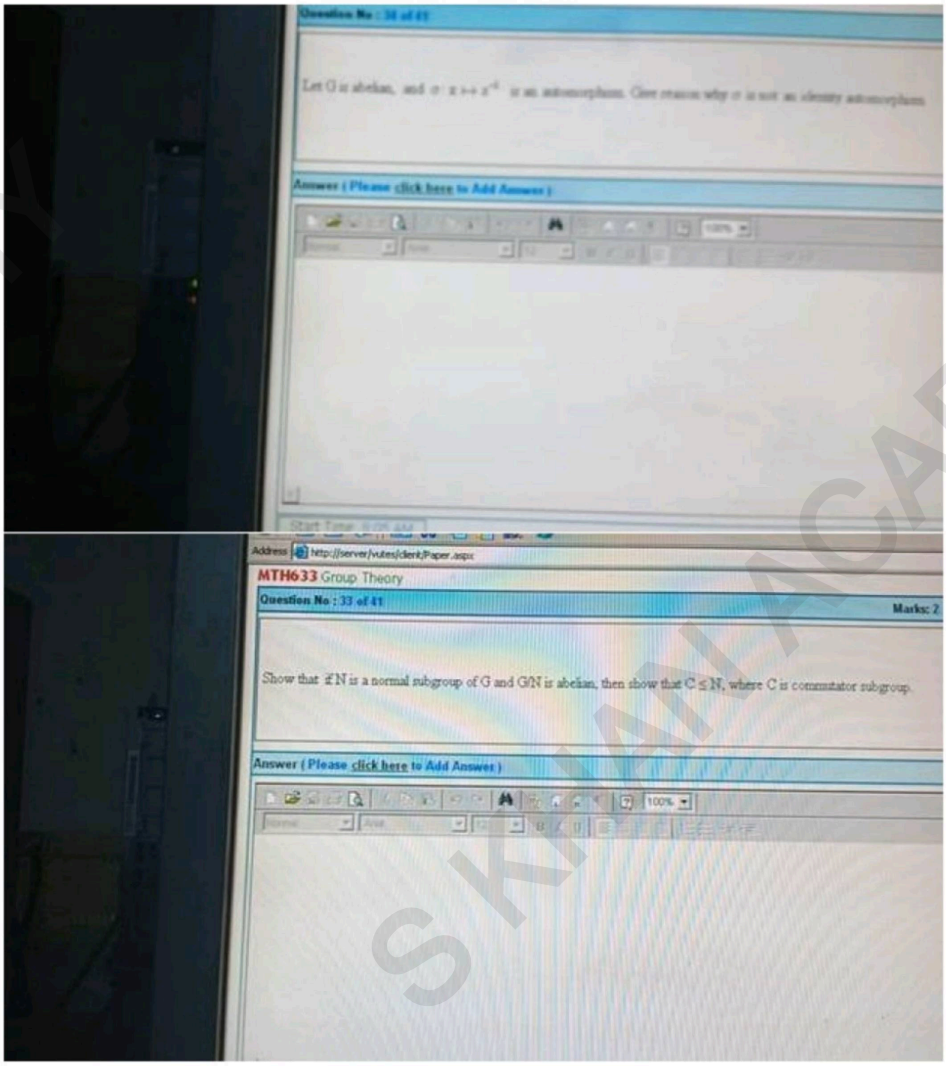

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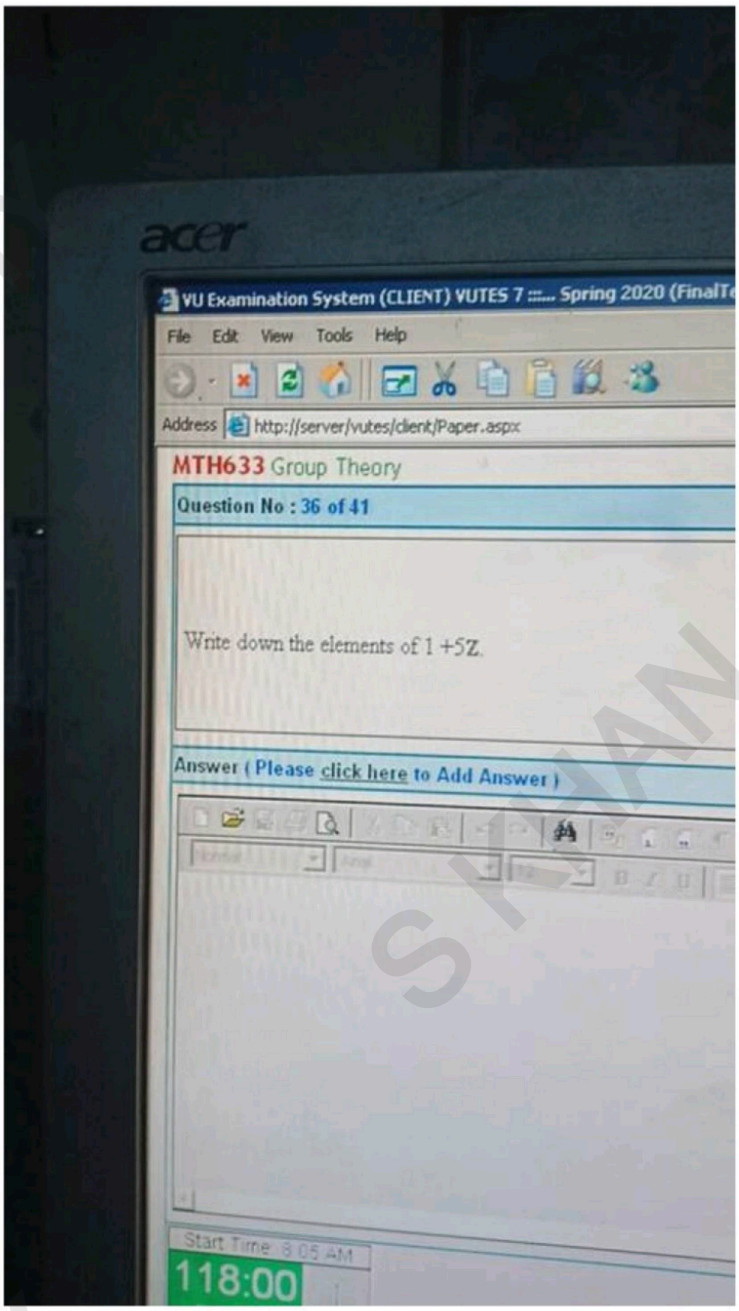

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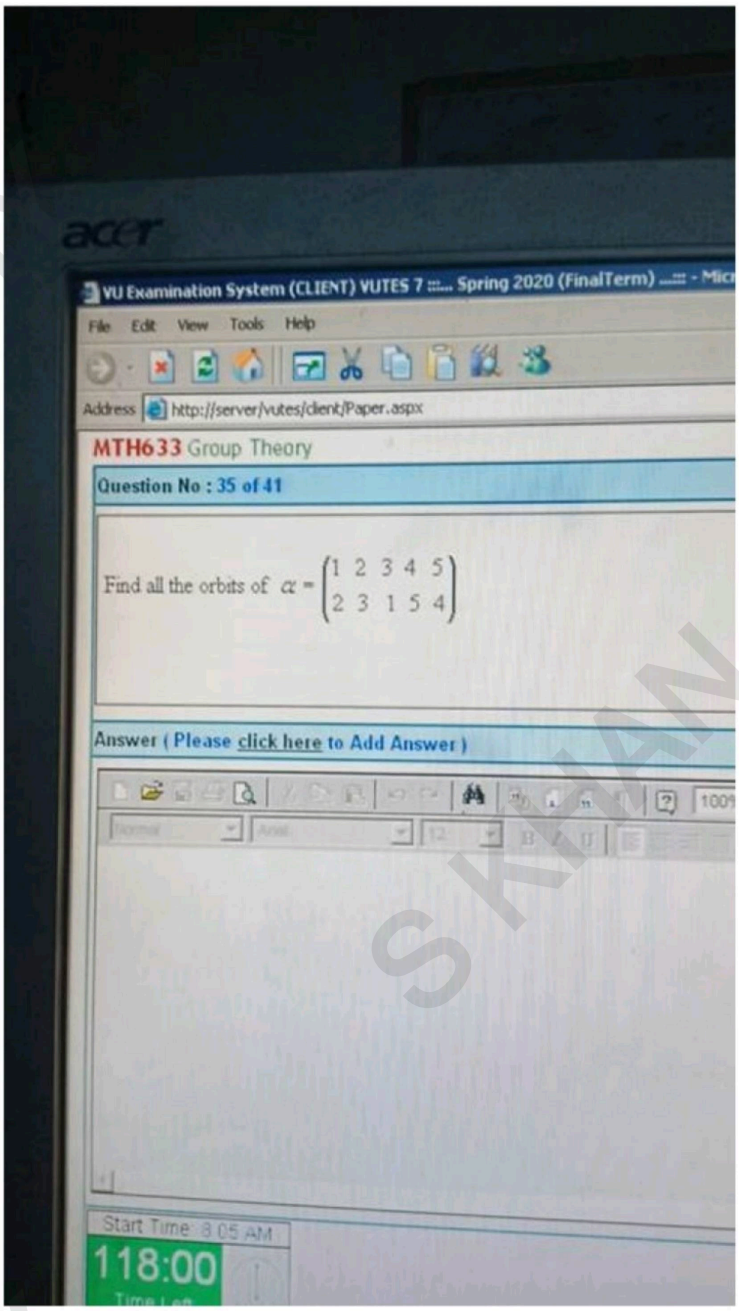

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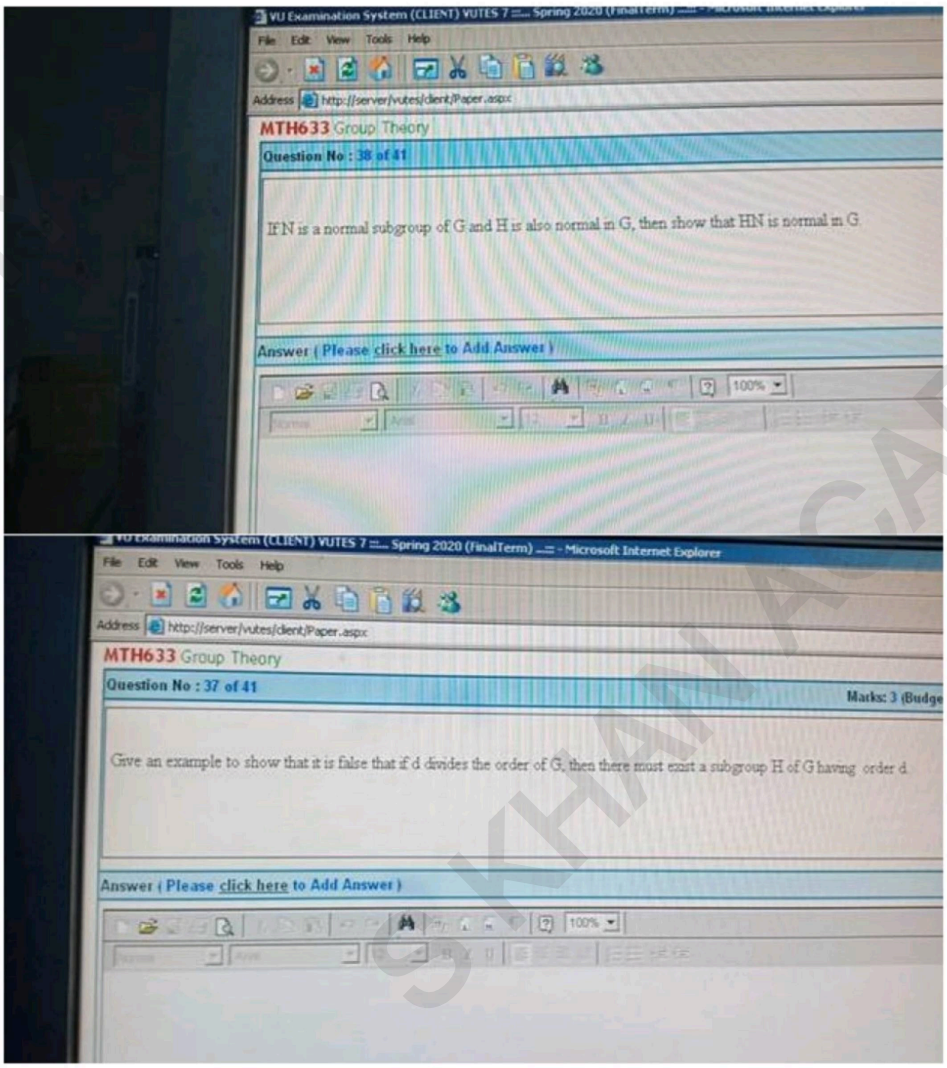

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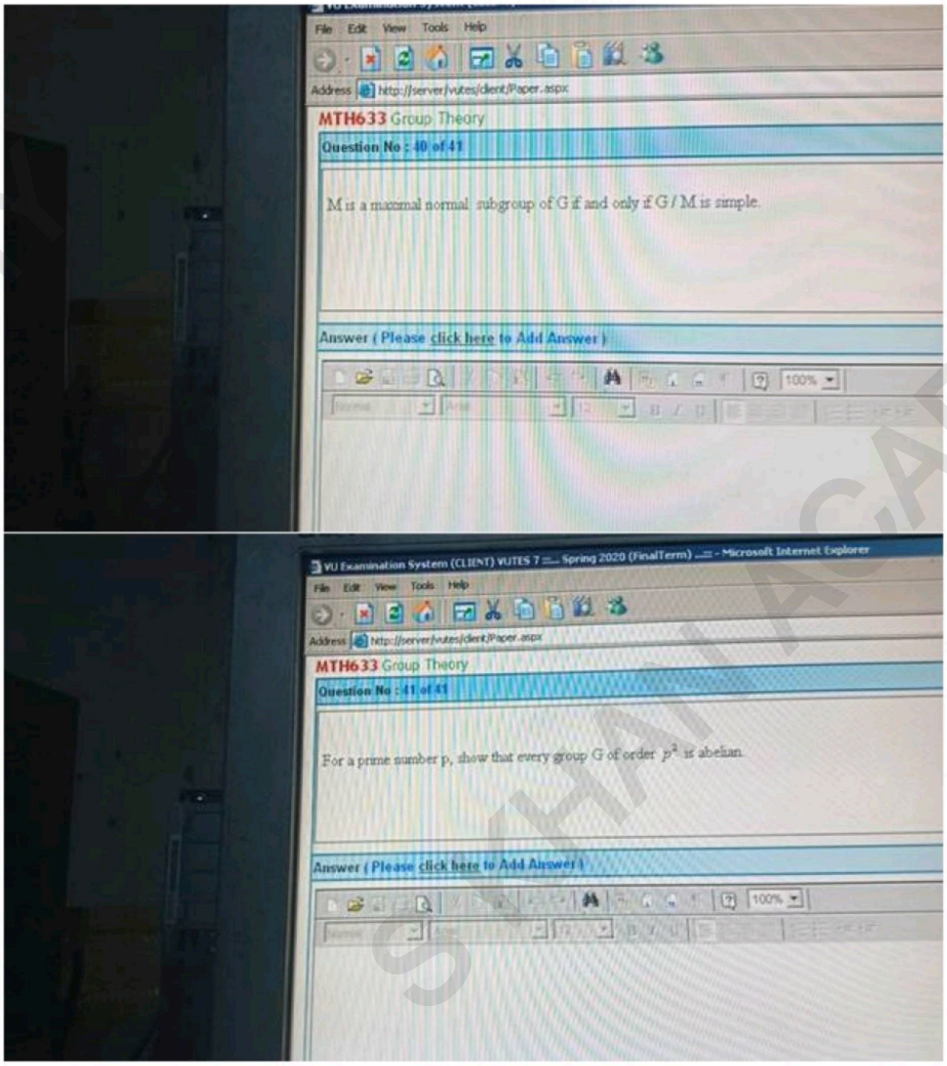

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